

# JESS Thermodynamic Database v8.9

## Germanium

24-Jan-23

| Reaction No. 4779                                                   |    |       |               |              |   |       |        |
|---------------------------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Ge+4 + 2<e-1> = Ge+2                                                |    |       |               |              |   |       |        |
| Note: Ge+4 is deprecated because does not exist in aqueous solution |    |       |               |              |   |       |        |
| No                                                                  | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0     | Inf. Dilution | E:0.0(0.1SD) | 0 | ENS   | 775    |
| 2                                                                   | 25 | 0     | Inf. Dilution | E:0.0(1SF)   | 0 | CRV   | 6330   |
| 3                                                                   | 25 | 0     | Inf. Dilution | E:0.0(1SF)   | 0 | CRV   | 22551  |
| 4                                                                   | 25 | 1     | H2SO4         | E:0.0(1SF)   | 0 | MEF   | 21221  |

| Reaction No. 4786                                      |    |       |               |                |   |       |        |
|--------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Ge+2 + 2<e-1> = Ge(s)                                  |    |       |               |                |   |       |        |
| Note: All secondary references based on unknown source |    |       |               |                |   |       |        |
| No                                                     | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                      | 25 | 0     | Inf. Dilution | E:0.23(0.01SD) | 3 | MDG   | 775    |
| 2                                                      | 25 | 0     | Inf. Dilution | E:0.24(2SF)    | 3 | CRV   | 6330   |
| 3                                                      | 25 | 0     | Inf. Dilution | E:0.247(3SF)   | 1 | CRV   | 22551  |
| 4                                                      | 25 | 0     | Inf. Dilution | E:0.231(3SF)   | 3 | EOD   | 31878  |
| 5                                                      | 25 | 1     | H2SO4         | E:0.24(2SF)    | 3 | MEF   | 21221  |

| Reaction No. 11875                                                 |    |       |          |                 |   |       |        |
|--------------------------------------------------------------------|----|-------|----------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1_Maltol-1> = Ge+4_Maltol-1(2)_OH-1(2) + 2<H2O> |    |       |          |                 |   |       |        |
| No                                                                 | t  | I-Str | Medium   | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                  | 20 | 0.1   | KNO3     | lgK:4.2(2SF)    | 4 | MGL   | 31905  |
| 2                                                                  | 25 | 0.5   | (H,Na)Cl | lgK:3.90(3SF)   | 4 | MSP   | 31893  |
| 3                                                                  | 25 | 0.5   | NaCl     | lgK:3.9(0.04SD) | 2 | MSP   | 999    |

| Reaction No. 11876                                               |    |       |        |                  |   |       |        |
|------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Ge+4_OH-1(4) + 3<H+1_Maltol-1> + H+1 = Ge+4_Maltol-1(3) + 4<H2O> |    |       |        |                  |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                | 20 | 0.1   | KNO3   | lgK:8.3(2SF)     | 4 | MGL   | 31905  |
| 2                                                                | 25 | 0.5   | HCl    | lgK:8.05(0.07SD) | 4 | MSP   | 31893  |
| 3                                                                | 25 | 0.5   | NaCl   | lgK:8.1(0.07SD)  | 2 | MSP   | 999    |

| Reaction No. 11918                                                                                 |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Ge+4\_OH-1(4)} + 2\text{<H+1\_Kojic-1>} = \text{Ge+4\_Kojic-1(2)\_OH-1(2)} + 2\text{<H2O>}$ |    |       |        |                 |   |       |        |
| No                                                                                                 | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                  | 20 | 0.1   | KNO3   | lgK:3.2(2SF)    | 4 | MGL   | 31905  |
| 2                                                                                                  | 25 | 0.5   | NaCl   | lgK:2.8(0.05SD) | 2 | MGL   | 999    |
| 3                                                                                                  | 25 | 0.5   | NaCl   | lgK:2.81(3SF)   | 4 | MSP   | 31976  |

| Reaction No. 12981                                                                                         |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Ge+4\_OH-1(4)} + 2\text{<H+1\_Tropolone-1>} = \text{Ge+4\_OH-1(2)\_Tropolone-1(2)} + 2\text{<H2O>}$ |    |       |        |                 |   |       |        |
| No                                                                                                         | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                          | 25 | 0.5   | NaCl   | lgK:8.0(0.05SD) | 2 | MSP   | 999    |
| 2                                                                                                          | 25 | 0.5   | NaCl   | lgK:8.03(3SF)   | 4 | MSP   | 31893  |

| Reaction No. 12982                                                                                             |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Ge+4\_OH-1(4)} + 3\text{<H+1\_Tropolone-1>} + \text{H+1} = \text{Ge+4\_Tropolone-1(3)} + 4\text{<H2O>}$ |    |       |        |                 |   |       |        |
| No                                                                                                             | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                              | 25 | 0.5   | NaCl   | lgK:13.3(0.1SD) | 4 | MSP   | 999    |
| 2                                                                                                              | 25 | 0.5   | NaCl   | lgK:13.3(0.1SD) | 4 | MSP   | 31893  |

| Reaction No. 13532                                                          |    |       |        |               |   |       |        |
|-----------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Ge+4\_Mandelic-1(3)} + \text{H+1} = \text{Ge+4\_H+1\_Mandelic-1(3)}$ |    |       |        |               |   |       |        |
| No                                                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0.1   | KCl    | lgK:3.83(3SF) | 2 | MGL   | 999    |
| 2                                                                           | 25 | 0.1   | KCl    | lgK:3.83(3SF) | 4 | MGL   | 3237   |

| Reaction No. 13533                                                                  |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Ge+4\_H+1\_Mandelic-1(3)} + \text{H+1} = \text{Ge+4\_H+1(2)\_Mandelic-1(3)}$ |    |       |        |               |   |       |        |
| No                                                                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 0.1   | KCl    | lgK:2.53(3SF) | 2 | MGL   | 999    |
| 2                                                                                   | 25 | 0.1   | KCl    | lgK:2.53(3SF) | 4 | MGL   | 3237   |

| Reaction No. 13726                                                                                 |    |       |        |               |   |       |        |
|----------------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Ge+4\_OH-1(4)} + 2\text{<H+1\_Oxine-1>} = \text{Ge+4\_OH-1(2)\_Oxine-1(2)} + 2\text{<H2O>}$ |    |       |        |               |   |       |        |
| No                                                                                                 | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                  | 25 | 0.5   | NaCl   | lgK:6.61(3SF) | 3 | MSP   | 999    |
| 2                                                                                                  | 25 | 0.5   | NaCl   | lgK:6.61(3SF) | 4 | MSP   | 31977  |

| Reaction No. 13776                                                                                   |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Ge+4\_OH-1(4)} + 2\text{<H+1\_Sulfox-2>} = \text{Ge+4\_OH-1(2)\_Sulfox-2(2)} + 2\text{<H2O>}$ |    |       |        |                |   |       |        |
| No                                                                                                   | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                    | 25 | 0.5   | NaCl   | lgK:6.55 (3SF) | 3 | MSP   | 999    |
| 2                                                                                                    | 25 | 0.5   | NaCl   | lgK:6.55 (3SF) | 4 | MSP   | 31977  |

| Reaction No. 13837                                                                                   |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Ge+4\_OH-1(4)} + 2\text{<H+1\_Ferron-2>} = \text{Ge+4\_OH-1(2)\_Ferron-2(2)} + 2\text{<H2O>}$ |    |       |        |                |   |       |        |
| No                                                                                                   | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                    | 25 | 0.5   | NaCl   | lgK:6.78 (3SF) | 3 | MSP   | 999    |
| 2                                                                                                    | 25 | 0.5   | NaCl   | lgK:6.78 (3SF) | 4 | MSP   | 31977  |

| Reaction No. 14005                                                                         |    |       |         |                |   |       |        |
|--------------------------------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| $\text{Ge+4\_OH-1(4)} + \text{H+1(3)\_EDTA-4} = \text{Ge+4\_OH-1\_EDTA-4} + 3\text{<H2O>}$ |    |       |         |                |   |       |        |
| Note: Complex initially Ge(OH)2HL, later Ge(OH)L; this form adopted in JESS                |    |       |         |                |   |       |        |
| No                                                                                         | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                          | 20 | 0.1   | KNO3    | lgK:4.57 (3SF) | 5 | MGL   | 3450   |
| 2                                                                                          | 25 | 0.1   | KClO4   | lgK:4.58 (3SF) | 5 | MPL   | 931    |
| Note (2): From 67Kon [31894]; thus complex written as Ge(OH)2HL                            |    |       |         |                |   |       |        |
| 3                                                                                          | 25 | 0.1   | KClO4   | lgK:4.58 (3SF) | 5 | MPL   | 31894  |
| Note (3): Complex written as Ge(OH)2HL in paper; modified later by authors                 |    |       |         |                |   |       |        |
| 4                                                                                          | 25 | 0.1   | Unknown | lgK:4.52 (3SF) | 4 | MGL   | 906    |
| 5                                                                                          | 25 | 0.1   | Unknown | lgK:4.52 (3SF) | 4 | MGL   | 31978  |

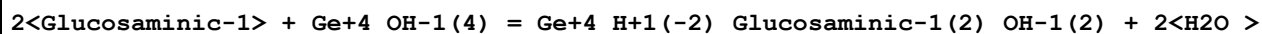
| Reaction No. 14006                                                                                           |    |       |        |                |   |       |        |
|--------------------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Glucosaminic-1} + \text{Ge+4\_OH-1(4)} = \text{GeO+2\_H+1(-1)\_Glucosaminic-1\_OH-1} + 2\text{<H2O>}$ |    |       |        |                |   |       |        |
| No                                                                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                            | 25 | 0.1   | NaClO4 | lgK:3.01 (3SF) | 5 | MGL   | 9211   |

| Reaction No. 14127                                                                                |    |       |          |                |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|----------|----------------|---|-------|--------|
| $\text{Ge+4\_OH-1(4)} + \text{H+1(3)\_HOEDTA-3} = \text{Ge+4\_H+1(-1)\_HOEDTA-3} + 4\text{<H2O>}$ |    |       |          |                |   |       |        |
| No                                                                                                | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 0.1   | NaClO4   | lgK:4.44 (3SF) | 0 | MGL   | 906    |
| 2                                                                                                 | 25 | 0.1   | NaClO4/m | lgK:4.44 (3SF) | 4 | MSP   | 6860   |

| Reaction No. 14389                                                                             |  |  |  |  |  |  |  |
|------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{Ge+4\_OH-1(4)} + \text{H+1(3)\_MeEDTA-4} = \text{Ge+4\_OH-1\_MeEDTA-4} + 3\text{<H2O>}$ |  |  |  |  |  |  |  |

| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4/m | lgK:4.88 (3SF) | 5 | MGL   | 6860   |

## Reaction No. 19788



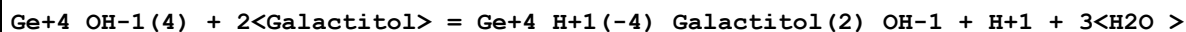
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:6.63 (0.01SD) | 5 | MGL   | 9211   |

## Reaction No. 21476



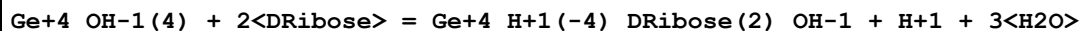
| No | t  | I-Str | Medium | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaCl   | lgK:-3.95 (0.01SD)   | 2 | MSC   | 999    |
| 2  | 25 | 0.5   | NaCl   | lgK:-3.945 (0.004SD) | 5 | MGL   | 7778   |

## Reaction No. 28563



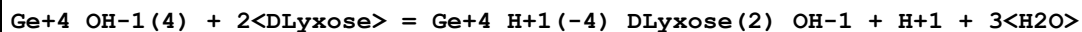
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3    | lgK:-3.88 (3SF) | 4 | MGL   | 31906  |
| 2  | 20 | 0.1   | Unknown | lgK:-3.88 (3SF) | 3 | CRV   | 2556   |
| 3  | 25 | 0.1   | KCl     | lgK:-3.87 (3SF) | 4 | MGL   | 31876  |
| 4  | 25 | 0.1   | Unknown | lgK:-3.87 (3SF) | 3 | CRV   | 2556   |

## Reaction No. 28569



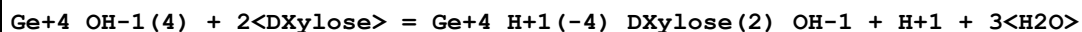
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:-2.46 (3SF) | 4 | CRV   | 2556   |

## Reaction No. 28572



| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:-3.94 (3SF) | 4 | CRV   | 2556   |

## Reaction No. 28576



| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl     | lgK:-5.200 (2SF) | 4 | MGL   | 31981  |
| 2  | 25 | 0.1   | Unknown | lgK:-5.20 (3SF)  | 3 | CRV   | 2556   |

| Reaction No. 28580                                                             |    |       |         |                  |   |       |        |
|--------------------------------------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<DArabinose> = Ge+4_H+1(-4)_DArabinose(2)_OH-1 + H+1 + 3<H2O > |    |       |         |                  |   |       |        |
| No                                                                             | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                              | 25 | 0.1   | KCl     | lgK:-5.065 (2SF) | 4 | MGL   | 31981  |
| 2                                                                              | 25 | 0.1   | Unknown | lgK:-5.07 (3SF)  | 3 | CRV   | 2556   |

| Reaction No. 28584                                                             |    |       |         |                  |   |       |        |
|--------------------------------------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<LArabinose> = Ge+4_H+1(-4)_LArabinose(2)_OH-1 + H+1 + 3<H2O > |    |       |         |                  |   |       |        |
| No                                                                             | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                              | 25 | 0.1   | KCl     | lgK:-4.955 (2SF) | 4 | MGL   | 31981  |
| 2                                                                              | 25 | 0.1   | Unknown | lgK:-4.96 (3SF)  | 3 | CRV   | 2556   |

| Reaction No. 28589                                                           |    |       |         |                 |   |       |        |
|------------------------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<DFructose> = Ge+4_H+1(-4)_DFructose(2)_OH-1 + H+1 + 3<H2O > |    |       |         |                 |   |       |        |
| No                                                                           | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.1   | KCl     | lgK:-3.10 (3SF) | 4 | MGL   | 31949  |
| 2                                                                            | 25 | 0.1   | Unknown | lgK:-3.10 (3SF) | 3 | CRV   | 2556   |

| Reaction No. 28592                                                         |    |       |         |                  |   |       |        |
|----------------------------------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<LSorbose> = Ge+4_H+1(-4)_LSorbose(2)_OH-1 + H+1 + 3<H2O > |    |       |         |                  |   |       |        |
| No                                                                         | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                          | 25 | 0.1   | KCl     | lgK:-3.235 (2SF) | 4 | MGL   | 31981  |
| 2                                                                          | 25 | 0.1   | Unknown | lgK:-3.24 (3SF)  | 3 | CRV   | 2556   |

| Reaction No. 28603                                                         |    |       |         |                  |   |       |        |
|----------------------------------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<DMannose> = Ge+4_H+1(-4)_DMannose(2)_OH-1 + H+1 + 3<H2O > |    |       |         |                  |   |       |        |
| No                                                                         | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                          | 25 | 0.1   | KCl     | lgK:-4.453 (2SF) | 4 | MGL   | 31873  |
| 2                                                                          | 25 | 0.1   | Unknown | lgK:-4.45 (3SF)  | 3 | CRV   | 2556   |

| Reaction No. 28606                                                             |    |       |         |                  |   |       |        |
|--------------------------------------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<DGalactose> = Ge+4_H+1(-4)_DGalactose(2)_OH-1 + H+1 + 3<H2O > |    |       |         |                  |   |       |        |
| No                                                                             | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                              | 25 | 0.1   | KCl     | lgK:-5.295 (2SF) | 4 | MGL   | 31873  |
| 2                                                                              | 25 | 0.1   | Unknown | lgK:-4.45 (3SF)  | 0 | CRV   | 2556   |
| Note (2): Error, this value is for DMannose                                    |    |       |         |                  |   |       |        |

| Reaction No. 31972                                            |    |       |        |               |   |       |        |
|---------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Ge+4_OH-1(2) + 2<BuXanthate-1> = Ge+4_BuXanthate-1(2)_OH-1(2) |    |       |        |               |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                             | 20 | 0.1   | KNO3   | lgK:8.82(3SF) | 5 | MSP   | 2261   |

| Reaction No. 32845                                                          |    |       |        |               |   |       |        |
|-----------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Ge+4_OH-1(4) + 3<H+1(2)_4NitrCat-2> = Ge+4_4NitrCat-2(3) + 2<H+1> + 4<H2O > |    |       |        |               |   |       |        |
| No                                                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0.1   | KCl    | lgK:3.90(3SF) | 3 | MSP   | 999    |
| 2                                                                           | 25 | 0.1   | KCl    | lgK:3.90(3SF) | 4 | MSP   | 31872  |

| Reaction No. 33581                                                             |    |       |         |               |   |       |        |
|--------------------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Ge+4_OH-1(4) + 3<H+1(4)_Gallic-4> = Ge+4_H+1(6)_Gallic-4(3) + 2<H+1> + 4<H2O > |    |       |         |               |   |       |        |
| No                                                                             | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                              | 25 | 0.1   | Unknown | lgK:-2.1(2SF) | 5 | CRV   | 2556   |

| Reaction No. 33582                                      |    |       |         |              |   |       |        |
|---------------------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| Ge+4_H+1(5)_Gallic-4(3) + H+1 = Ge+4_H+1(6)_Gallic-4(3) |    |       |         |              |   |       |        |
| No                                                      | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | Unknown | lgK:4.6(2SF) | 5 | CRV   | 2556   |

| Reaction No. 33583                                      |    |       |         |              |   |       |        |
|---------------------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| Ge+4_H+1(4)_Gallic-4(3) + H+1 = Ge+4_H+1(5)_Gallic-4(3) |    |       |         |              |   |       |        |
| No                                                      | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | Unknown | lgK:5.1(2SF) | 5 | CRV   | 2556   |

| Reaction No. 33584                                      |    |       |         |              |   |       |        |
|---------------------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| Ge+4_H+1(3)_Gallic-4(3) + H+1 = Ge+4_H+1(4)_Gallic-4(3) |    |       |         |              |   |       |        |
| No                                                      | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | Unknown | lgK:5.6(2SF) | 5 | CRV   | 2556   |

| Reaction No. 34511                                        |    |       |               |               |   |       |        |
|-----------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Ge+4_OH-1(4) + H+1(2)_NTA-3 = Ge+4_OH-1(2)_NTA-3 + 2<H2O> |    |       |               |               |   |       |        |
| No                                                        | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                         | 20 | 0.1   | KNO3          | lgK:4.42(3SF) | 5 | MGL   | 3450   |
| 2                                                         | 25 | 0     | Inf. Dilution | lgK:4.5(2SF)  | 1 | ESC   |        |

| Reaction No. 34512 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Ge+4_OH-1(4) + H+1(3)_TriEtOlAm = Ge+4_TriEtOlAm_OH-1 + 3<H2O> |    |       |        |               |   |       |        |
|----------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| No                                                             | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                              | 20 | 0.1   | KNO3   | lgK:5.26(3SF) | 5 | MGL   | 3450   |

| Reaction No. 34708                                                               |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1_2SHetol-1> = Ge+4_H+1(-2)_2SHetol-1(2)_OH-1 + H+1 + 3<H2O > |    |       |        |                 |   |       |        |
| No                                                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                | 25 | 0.1   | KCl    | lgK:-4.22(3SF)  | 3 | MGL   | 931    |
| 2                                                                                | 25 | 0.1   | KCl    | lgK:-4.215(2SF) | 4 | MGL   | 31982  |

| Reaction No. 34720                                                                        |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| GeO+2_OH-1(3) + 2<Dfructose> = GeO+2_H+1(-3)_Dfructose(2) + 3<H2O>                        |    |       |        |               |   |       |        |
| No                                                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0.1   | KCl    | lgK:5.48(3SF) | 0 | MGL   | 931    |
| Note (1): Same source rn. 28589 but written otherwise (GeO(OH)3 reactant instead Ge(OH)4) |    |       |        |               |   |       |        |

| Reaction No. 34726                                                                        |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| GeO+2_OH-1(3) + 2<DGalactose> = GeO+2_H+1(-3)_DGalactose(2) + 3<H2O>                      |    |       |        |               |   |       |        |
| No                                                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0.1   | KCl    | lgK:3.29(3SF) | 0 | MGL   | 931    |
| Note (1): Same source rn. 28606 but written otherwise (GeO(OH)3 reactant instead Ge(OH)4) |    |       |        |               |   |       |        |

| Reaction No. 34733                                                                        |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| GeO+2_OH-1(3) + 2<DGlucose> = GeO+2_H+1(-3)_DGlucose(2) + 3<H2O>                          |    |       |        |               |   |       |        |
| No                                                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0.1   | KCl    | lgK:3.46(3SF) | 0 | MGL   | 931    |
| Note (1): Same source rn. 80927 but written otherwise (GeO(OH)3 reactant instead Ge(OH)4) |    |       |        |               |   |       |        |

| Reaction No. 34741                                                                        |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| GeO+2_OH-1(3) + 2<DMannose> = GeO+2_H+1(-3)_DMannose(2) + 3<H2O>                          |    |       |        |               |   |       |        |
| No                                                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0.1   | KCl    | lgK:4.13(3SF) | 0 | MGL   | 931    |
| Note (1): Same source rn. 28603 but written otherwise (GeO(OH)3 reactant instead Ge(OH)4) |    |       |        |               |   |       |        |

|                    |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
| Reaction No. 37267 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|

| Ge+4_OH-1(4) + 2<H+1_ClKojic-1> = Ge+4_ClKojic-1(2)_OH-1(2) + 2<H2O> |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| No                                                                   | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0.5   | NaCl   | lgK:2.3(0.05SD) | 3 | MSP   | 931    |
| 2                                                                    | 25 | 0.5   | NaCl   | lgK:2.33(3SF)   | 4 | MSP   | 31976  |

| Reaction No. 37342                                                                  |    |       |        |              |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1_Lawsone-1> = Ge+4_Lawsone-1(2)_OH-1(2) + 2<H2O>                |    |       |        |              |   |       |        |
| No                                                                                  | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 0.5   | NaCl   | lgK:3.0(2SF) | 0 | MSP   | 931    |
| 2                                                                                   | 25 | 0.5   | NaCl   | lgK:3.0(2SF) | 0 | MSP   | 31893  |
| Note (2): The value could not be determined and the authors just gave a limit: <3.0 |    |       |        |              |   |       |        |

| Reaction No. 40764                                                                        |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| GeO+2_OH-1(3) + 2<LRhamnose> = GeO+2_H+1(-3)_LRhamnose(2) + 3<H2O>                        |    |       |        |               |   |       |        |
| No                                                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0.1   | KCl    | lgK:3.24(3SF) | 0 | MGL   | 931    |
| Note (1): Same source rn. 80929 but written otherwise (GeO(OH)3 reactant instead Ge(OH)4) |    |       |        |               |   |       |        |

| Reaction No. 40767                                                                        |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| GeO+2_OH-1(3) + 2<LSorbose> = GeO+2_H+1(-3)_LSorbose(2) + 3<H2O>                          |    |       |        |               |   |       |        |
| No                                                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0.1   | KCl    | lgK:5.35(3SF) | 0 | MGL   | 931    |
| Note (1): Same source rn. 28592 but written otherwise (GeO(OH)3 reactant instead Ge(OH)4) |    |       |        |               |   |       |        |

| Reaction No. 41152                                                               |    |       |        |              |   |       |        |
|----------------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1_57DiCl8OHQuin-1> = Ge+4_57DiCl8OHQuin-1(2)_OH-1(2) + 2<H2O> |    |       |        |              |   |       |        |
| No                                                                               | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                | 25 | 0.5   | NaCl   | lgK:6.7(2SF) | 3 | MSP   | 931    |
| 2                                                                                | 25 | 0.5   | NaCl   | lgK:6.7(2SF) | 4 | MSP   | 31977  |

| Reaction No. 43245                                                                     |    |       |        |               |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Ge+4_OH-1(2) + H+1_Tartaric-2 = Ge+4_H+1(-1)_OH-1(2)_Tartaric-2 + 2<H+1>               |    |       |        |               |   |       |        |
| Note: Ge(OH)2 deprecated, existence very much doubtful, even more in these experiments |    |       |        |               |   |       |        |
| No                                                                                     | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                      | 25 | 0.1   | NaNO3  | lgK:4.13(3SF) | 0 | MSP   | 906    |



|                                                                                                                                    |    |     |       |                |   |     |       |
|------------------------------------------------------------------------------------------------------------------------------------|----|-----|-------|----------------|---|-----|-------|
| Note (1): Reaction written: $\text{Ge}(\text{OH})_2 + \text{H}_2\text{Tartaric} = \text{GeH}_2(\text{OH})_2\text{Tartaric}$ in 906 |    |     |       |                |   |     |       |
| 2                                                                                                                                  | 25 | 0.1 | NaNO3 | lgK:4.13 (3SF) | 0 | MSP | 31901 |

| Reaction No. 43733                                                                                            |    |       |        |                |   |       |        |
|---------------------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Ge+4\_OH-1(2)} + \text{H+1(3)\_Citric-3} = \text{Ge+4\_H+1(3)\_OH-1(2)\_Citric-3}$                     |    |       |        |                |   |       |        |
| Note: $\text{Ge}(\text{OH})_2$ deprecated, existence very much doubtful in general, sure in these experiments |    |       |        |                |   |       |        |
| No                                                                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                             | 25 | 0.1   | NaNO3  | lgK:2.01 (3SF) | 0 | MSP   | 906    |
| Note (1): Unclear but in original reference [31901], reaction is for ML2 formation                            |    |       |        |                |   |       |        |

| Reaction No. 44092                                                           |    |       |        |                   |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $3<35\text{DiNitrCat-2}> + \text{Ge+4} = \text{Ge+4\_35DiNitrCat-2(3)}$      |    |       |        |                   |   |       |        |
| Note: $\text{Ge+4}$ is deprecated because does not exist in aqueous solution |    |       |        |                   |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.1   | NaClO4 | lgK:47.9 (0.08SD) | 0 | MSP   | 906    |
| 2                                                                            | 25 | 0.1   | NaClO4 | lgK:47.85 (4SF)   | 0 | MSP   | 31899  |

| Reaction No. 44096                                                           |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $3<3\text{NitrCat-2}> + \text{Ge+4} = \text{Ge+4\_3NitrCat-2(3)}$            |    |       |        |                 |   |       |        |
| Note: $\text{Ge+4}$ is deprecated because does not exist in aqueous solution |    |       |        |                 |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.1   | NaClO4 | lgK:59.59 (4SF) | 0 | MSP   | 906    |
| 2                                                                            | 25 | 0.1   | NaClO4 | lgK:59.59 (4SF) | 0 | MSP   | 31950  |

| Reaction No. 45090                                                            |    |       |          |                |   |       |        |
|-------------------------------------------------------------------------------|----|-------|----------|----------------|---|-------|--------|
| $\text{H+1} + \text{Ge+4\_OH-1\_MeEDTA-4} = \text{Ge+4\_H+1\_OH-1\_MeEDTA-4}$ |    |       |          |                |   |       |        |
| No                                                                            | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                             | 25 | 0.1   | NaClO4/m | lgK:2.50 (3SF) | 5 | MGL   | 6860   |
| 2                                                                             | 25 | 0.1   | Unknown  | lgK:2.50 (3SF) | 3 | CRV   | 552    |

| Reaction No. 45091                                                                                          |    |       |          |                |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|----------|----------------|---|-------|--------|
| $\text{H+1(4)\_MeEDTA-4} + \text{Ge+4\_OH-1(4)} = \text{Ge+4\_H+1\_OH-1\_MeEDTA-4} + 3<\text{H}_2\text{O}>$ |    |       |          |                |   |       |        |
| No                                                                                                          | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                           | 25 | 0.1   | NaClO4/m | lgK:4.78 (3SF) | 5 | MGL   | 6860   |
| 2                                                                                                           | 25 | 0.1   | Unknown  | lgK:4.78 (3SF) | 3 | CRV   | 552    |

| Reaction No. 45230                                                  |    |       |             |                 |   |       |        |
|---------------------------------------------------------------------|----|-------|-------------|-----------------|---|-------|--------|
| 3<AlizarinRedS-3> + Ge+4 = Ge+4_AlizarinRedS-3(3)                   |    |       |             |                 |   |       |        |
| Note: Ge+4 is deprecated because does not exist in aqueous solution |    |       |             |                 |   |       |        |
| No                                                                  | t  | I-Str | Medium      | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.1   | Acetic acid | lgK:52.80 (4SF) | 0 | MSP   | 906    |

| Reaction No. 46454                    |    |       |         |                 |   |       |        |
|---------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| 2<Cl-1> + Ge+4_Cl-1(4) = Ge+4_Cl-1(6) |    |       |         |                 |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 23 | 0     | Unknown | lgK:-5.06 (3SF) | 1 | MSP   | 31970  |

| Reaction No. 54155              |    |       |               |                |   |       |        |
|---------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Ge+4_F-1(5) + F-1 = Ge+4_F-1(6) |    |       |               |                |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                               | 23 | 0     | Unknown       | lgK:3.21 (3SF) | 1 | MIX   | 31921  |
| 2                               | 23 | 0.5   | HCl           | lgK:3.86 (3SF) | 1 | MAX   | 931    |
| 3                               | 23 | 0.5   | KCl           | lgK:3.86 (3SF) | 1 | MIX   | 31920  |
| 4                               | 25 | 0     | Inf. Dilution | lgK:3.56 (3SF) | 1 | EAE   |        |

| Reaction No. 54432 |    |       |               |                         |   |       |        |
|--------------------|----|-------|---------------|-------------------------|---|-------|--------|
| Ge(s) = Ge(g)      |    |       |               |                         |   |       |        |
| No                 | t  | I-Str | Medium        | Value (Dev)             | W | Tech. | Ref. # |
| 1                  | 25 | 0     | Inf. Dilution | dG:331.21 (3.000SD) kJ  | 6 | CRV   | 14923  |
| 2                  | 25 | 0     | Inf. Dilution | dH:372. (3.MD) kJ       | 6 | CRV   | 8746   |
| 3                  | 25 | 0     | Inf. Dilution | dH:372.00 (3.000SD) kJ  | 6 | CRV   | 14923  |
| 4                  | 25 | 0     | Inf. Dilution | Cp:30.733 (0.001SD) J/K | 5 | CRV   | 27292  |

| Reaction No. 54447                 |    |       |               |                 |   |       |        |
|------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| H+1 + GeO2_OH-1(2) = GeO+2_OH-1(3) |    |       |               |                 |   |       |        |
| No                                 | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                  | 12 | 2     | KCl           | lgK:12.7 (3SF)  | 1 | MGL   | 140    |
| 2                                  | 12 | 2     | KCl           | lgK:12.7 (3SF)  | 2 | MGL   | 19840  |
| 3                                  | 20 | 0     | Self Medium   | lgK:12.7 (3SF)  | 2 | MHY   | 31972  |
| 4                                  | 20 |       |               | lgK:12.72 (4SF) | 0 | MHY   | 140    |
| 5                                  | 25 | 0     | Inf. Dilution | lgK:12.59 (4SF) | 4 | CRV   | 403    |
| 6                                  | 25 | 0     | Inf. Dilution | lgK:12.3 (3SF)  | 0 | CRV   | 6330   |
| 7                                  | 25 | 0     | Inf. Dilution | lgK:13. (2SF)   | 0 | ENS   | 7276   |

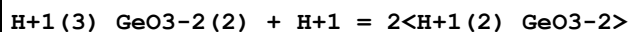
|                                    |    |     |        |                   |   |     |       |
|------------------------------------|----|-----|--------|-------------------|---|-----|-------|
| 8                                  | 25 | 0.5 | NaCl   | lgK:12.24(4SF)    | 2 | MGL | 140   |
| 9                                  | 25 | 3   | Na(Cl) | lgK:12.43(0.03SD) | 5 | MHY | 22980 |
| 10                                 | 25 | 3   | NaCl   | lgK:12.43(4SF)    | 2 | MHY | 140   |
| 11                                 | 32 | 5   | Na2SO4 | lgK:12.3(3SF)     | 1 | MHY | 31968 |
| Note (11): I is saturated, not 5 M |    |     |        |                   |   |     |       |
| 12                                 | 32 |     |        | lgK:12.3(3SF)     | 0 | MFP | 140   |

| Reaction No. 54448                 |    |        |               |                |   |       |        |
|------------------------------------|----|--------|---------------|----------------|---|-------|--------|
| GeO+2_OH-1(3) + H+1 = Ge+4_OH-1(4) |    |        |               |                |   |       |        |
| No                                 | t  | I-Str  | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                  | 10 | 0      | Inf. Dilution | lgK:9.045(4SF) | 3 | MGL   | 31870  |
| 2                                  | 10 | 0.6326 | KCl           | lgK:8.839(4SF) | 3 | MGL   | 31870  |
| 3                                  | 12 | 2      | KCl           | lgK:9.1(2SF)   | 3 | MGL   | 19840  |
| 4                                  | 15 | 0      | Inf. Dilution | lgK:8.980(4SF) | 3 | MGL   | 31870  |
| 5                                  | 15 | 0.1998 | KCl           | lgK:8.880(4SF) | 3 | MGL   | 31870  |
| 6                                  | 15 | 0.316  | KCl           | lgK:8.864(4SF) | 3 | MGL   | 31870  |
| 7                                  | 18 | 0      | Self Medium   | lgK:6.92(3SF)  | 0 | MCN   | 31960  |
| 8                                  | 18 | 0.5    | Na2SO4        | lgK:9.1(2SF)   | 3 | MGL   | 31966  |
| 9                                  | 18 |        |               | lgK:6.92(3SF)  | 0 | MCN   | 140    |
| 10                                 | 20 | 0      | Inf. Dilution | lgK:8.920(4SF) | 3 | MGL   | 31870  |
| 11                                 | 20 | 0      | Unknown       | lgK:7.30(3SF)  | 0 | MCN   | 31869  |
| 12                                 | 20 | 0      | Unknown       | lgK:8.6(2SF)   | 1 | MHY   | 31972  |
| 13                                 | 20 | 0.1    | KNO3          | lgK:9.27(3SF)  | 4 | MGL   | 31905  |
| 14                                 | 20 | 0.6318 | KCl           | lgK:8.635(4SF) | 3 | MGL   | 31870  |
| 15                                 | 20 |        |               | lgK:7.30(3SF)  | 0 | MCN   | 140    |
| 16                                 | 20 |        |               | lgK:8.59(3SF)  | 0 | MHY   | 140    |
| 17                                 | 21 | 0      | Inf. Dilution | lgK:9.37(3SF)  | 5 | MGL   | 8763   |
| 18                                 | 25 | 0      | Inf. Dilution | lgK:9.4(2SF)   | 1 | ELC   |        |
| 19                                 | 25 | 0      | Inf. Dilution | lgK:9.0(2SF)   | 0 | MCN   | 140    |
| 20                                 | 25 | 0      | Inf. Dilution | lgK:9.4(2SF)   | 0 | MCO   | 140    |
| 21                                 | 25 | 0      | Inf. Dilution | lgK:8.82(3SF)  | 0 | MDG   | 140    |
| 22                                 | 25 | 0      | Inf. Dilution | lgK:9.3(0.2SD) | 5 | CRV   | 403    |
| 23                                 | 25 | 0      | Inf. Dilution | lgK:9.01(3SF)  | 0 | CRV   | 6330   |
| 24                                 | 25 | 0      | Inf. Dilution | lgK:8.59(3SF)  | 0 | ENS   | 7276   |
| 25                                 | 25 | 0      | Inf. Dilution | lgK:9.32(3SF)  | 5 | MGL   | 8763   |
| 26                                 | 25 | 0      | Inf. Dilution | lgK:8.730(4SF) | 3 | MGL   | 31870  |

|                                                         |     |         |                    |                  |   |     |       |
|---------------------------------------------------------|-----|---------|--------------------|------------------|---|-----|-------|
| 27                                                      | 25  | 0       | Self Medium        | lgK:11.6(3SF)    | 0 | MCN | 31971 |
| 28                                                      | 25  | 0       | Unknown            | lgK:8.8(2SF)     | 1 | MCN | 18662 |
| Note (28): Average of MCN and colorimetric measurements |     |         |                    |                  |   |     |       |
| 29                                                      | 25  | 0.0102  | KCl                | lgK:8.722(4SF)   | 3 | MGL | 31870 |
| 30                                                      | 25  | 0.1     | KCl                | lgK:9.26(0.03SD) | 5 | MGL | 31907 |
| 31                                                      | 25  | 0.1     | NaClO <sub>4</sub> | lgK:9.16(0.01SD) | 5 | MGL | 9211  |
| 32                                                      | 25  | 0.1751  | KCl                | lgK:8.632(4SF)   | 3 | MGL | 31870 |
| 33                                                      | 25  | 0.4558  | KCl                | lgK:8.526(4SF)   | 3 | MGL | 31870 |
| 34                                                      | 25  | 0.65    | KCl                | lgK:8.563(4SF)   | 3 | MGL | 31870 |
| 35                                                      | 25  | 0.9998  | KCl                | lgK:8.571(4SF)   | 3 | MGL | 31870 |
| 36                                                      | 25  | 1.231   | KCl                | lgK:8.751(4SF)   | 3 | MGL | 31870 |
| 37                                                      | 25  |         |                    | lgK:8.75(3SF)    | 0 | MCN | 140   |
| 38                                                      | 30  | 0       | Inf. Dilution      | lgK:8.615(4SF)   | 3 | MGL | 31870 |
| 39                                                      | 30  | 0.06352 | KCl                | lgK:8.378(4SF)   | 3 | MGL | 31870 |
| 40                                                      | 31  | 0       | Inf. Dilution      | lgK:9.26(3SF)    | 5 | MGL | 8763  |
| 41                                                      | 35  | 0       | Inf. Dilution      | lgK:8.450(4SF)   | 3 | MGL | 31870 |
| 42                                                      | 35  | 0.1987  | KCl                | lgK:8.311(4SF)   | 3 | MGL | 31870 |
| 43                                                      | 35  | 0.3158  | KCl                | lgK:8.204(4SF)   | 3 | MGL | 31870 |
| 44                                                      | 35  | 0.6363  | KCl                | lgK:8.208(4SF)   | 3 | MGL | 31870 |
| 45                                                      | 40  | 0       | Inf. Dilution      | lgK:8.175(4SF)   | 3 | MGL | 31870 |
| 46                                                      | 40  | 0.6296  | KCl                | lgK:7.827(4SF)   | 3 | MGL | 31870 |
| 47                                                      | 40  | 0.6356  | KCl                | lgK:7.913(4SF)   | 3 | MGL | 31870 |
| 48                                                      | 45  | 0       | Inf. Dilution      | lgK:7.900(4SF)   | 3 | MGL | 31870 |
| 49                                                      | 45  | 0.199   | KCl                | lgK:7.747(4SF)   | 3 | MGL | 31870 |
| 50                                                      | 45  | 0.6291  | KCl                | lgK:7.650(4SF)   | 3 | MGL | 31870 |
| 51                                                      | 45  | 0.6294  | KCl                | lgK:7.573(4SF)   | 3 | MGL | 31870 |
| 52                                                      | 50  | 0       | Inf. Dilution      | lgK:8.92(3SF)    | 5 | MGL | 8763  |
| 53                                                      | 60  | 0       | Inf. Dilution      | lgK:8.84(3SF)    | 5 | MGL | 8763  |
| 54                                                      | 75  | 0       | Inf. Dilution      | lgK:8.70(3SF)    | 5 | MGL | 8763  |
| 55                                                      | 90  | 0       | Inf. Dilution      | lgK:8.65(3SF)    | 5 | MGL | 8763  |
| 56                                                      | 90  | 0       | Inf. Dilution      | lgK:8.63(3SF)    | 5 | MSL | 8763  |
| 57                                                      | 100 | 0       | Inf. Dilution      | lgK:8.55(3SF)    | 5 | MGL | 8763  |
| 58                                                      | 125 | 0       | Inf. Dilution      | lgK:8.48(3SF)    | 5 | MGL | 8763  |
| 59                                                      | 150 | 0       | Inf. Dilution      | lgK:8.41(3SF)    | 5 | MGL | 8763  |
| 60                                                      | 175 | 0       | Inf. Dilution      | lgK:8.35(3SF)    | 5 | MGL | 8763  |
| 61                                                      | 190 | 0       | Inf. Dilution      | lgK:8.34(3SF)    | 5 | MGL | 8763  |

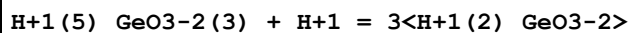
|    |     |   |               |                |   |     |      |
|----|-----|---|---------------|----------------|---|-----|------|
| 62 | 200 | 0 | Inf. Dilution | lgK:8.37 (3SF) | 5 | MGL | 8763 |
|----|-----|---|---------------|----------------|---|-----|------|

## Reaction No. 54449



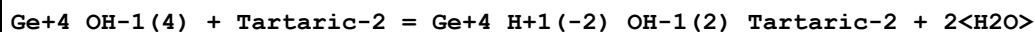
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaCl   | lgK:-4.3 (2SF) | 4 | MGL   | 140    |

## Reaction No. 54450



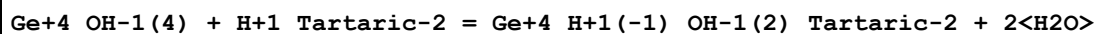
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaCl   | lgK:-3.64 (3SF) | 4 | MGL   | 140    |

## Reaction No. 54451



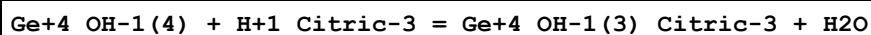
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaOH   | lgK:2.51 (3SF) | 4 | MSL   | 8740   |

## Reaction No. 54452



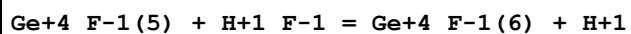
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaOH   | lgK:2.70 (3SF) | 4 | MSL   | 8740   |

## Reaction No. 54487



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaCl/m        | lgK:2.36 (0.30SD) | 5 | MSL   | 8740   |
| 2  | 90 | 0.1   | NaCl/m        | lgK:1.59 (0.10SD) | 5 | MSL   | 8740   |
| 3  | 25 | 0     | Inf. Dilution | dH:-5.87 (3SF)    | 4 | MTD   | 8740   |

## Reaction No. 54488



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 0  | 0.4   | NaCl   | lgK:0.66 (2SF) | 2 | MAN   | 31888  |
| 2  | 10 | 0.4   | NaCl   | lgK:0.52 (2SF) | 2 | MAN   | 31888  |
| 3  | 20 | 0.4   | NaCl   | lgK:0.58 (2SF) | 0 | MAN   | 437    |
| 4  | 20 | 0.4   | NaCl   | lgK:0.58 (2SF) | 2 | MSC   | 31888  |
| 5  | 30 | 0.4   | NaCl   | lgK:0.50 (2SF) | 2 | MAN   | 31888  |

|   |    |     |      |                |   |     |       |
|---|----|-----|------|----------------|---|-----|-------|
| 6 | 40 | 0.4 | NaCl | lgK:0.42 (2SF) | 2 | MAN | 31888 |
| 7 | 50 | 0.4 | NaCl | lgK:0.34 (2SF) | 2 | MAN | 31888 |

| Reaction No. 54627                                                                                                                       |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{GeO}_2(\text{hexag},s) + 2\langle\text{H}+1\rangle + 2\langle\text{e}-1\rangle = \text{GeO}(\text{black},s) + \text{H}_2\text{O}$ |    |       |               |                |   |       |        |
| Note: Polymorph of GeO <sub>2</sub> not identified in original source [52JoL]                                                            |    |       |               |                |   |       |        |
| No                                                                                                                                       | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                        | 25 | 0     | Unknown       | lgK:-4.0 (2SF) | 1 | ENS   | 140    |
| 2                                                                                                                                        | 25 | 0     | Inf. Dilution | E:-0.118 (3SF) | 3 | CRV   | 6330   |
| 3                                                                                                                                        | 25 | 0     | Inf. Dilution | E:-0.132 (3SF) | 0 | CRV   | 22551  |
| 4                                                                                                                                        | 25 | 0     | Unknown       | E:-0.118 (3SF) | 3 | MEF   | 8052   |
| Note (4): GeO <sub>2</sub> polymorph unknown, probably hexag                                                                             |    |       |               |                |   |       |        |

| Reaction No. 54628                                                                                                                        |    |       |               |                |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{GeO}_2(\text{hexag},s) + 2\langle\text{H}+1\rangle + 2\langle\text{e}-1\rangle = \text{GeO}(\text{yellow},s) + \text{H}_2\text{O}$ |    |       |               |                |   |       |        |
| No                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:-9.1 (2SF) | 1 | ENS   | 140    |

| Reaction No. 54629                                                                                                                       |    |       |               |                  |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{GeO}_2(\text{hexag},s) + 4\langle\text{H}+1\rangle + 2\langle\text{e}-1\rangle = \text{Ge}+2 + 2\langle\text{H}_2\text{O}\rangle$ |    |       |               |                  |   |       |        |
| No                                                                                                                                       | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                        | 25 | 0     | Inf. Dilution | lgK:-9.30 (3SF)  | 1 | ELC   |        |
| 2                                                                                                                                        | 25 | 0     | Inf. Dilution | E:-0.25 (0.01SD) | 2 | MDG   | 775    |

| Reaction No. 54630                                                                                                                        |    |       |               |                  |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{GeO}_2(\text{tetrag},s) + 4\langle\text{H}+1\rangle + 2\langle\text{e}-1\rangle = \text{Ge}+2 + 2\langle\text{H}_2\text{O}\rangle$ |    |       |               |                  |   |       |        |
| No                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:-12.46 (4SF) | 1 | ELC   |        |
| 2                                                                                                                                         | 25 | 0     | Inf. Dilution | E:-0.34 (0.01SD) | 2 | MDG   | 775    |

| Reaction No. 54631                                                                                                                        |    |       |               |                  |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{GeO}_2(\text{hexag},s) + 4\langle\text{e}-1\rangle + 4\langle\text{H}+1\rangle = \text{Ge}(s) + 2\langle\text{H}_2\text{O}\rangle$ |    |       |               |                  |   |       |        |
| No                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:-0.91 (2SF)  | 1 | ELC   |        |
| 2                                                                                                                                         | 25 | 0     | Inf. Dilution | E:-0.01 (0.01SD) | 2 | MDG   | 775    |
| 3                                                                                                                                         | 25 | 0     | Inf. Dilution | E:-0.019 (2SF)   | 0 | CRV   | 22551  |
| 4                                                                                                                                         | 25 | 0     | Inf. Dilution | E:-0.009 (1SF)   | 3 | EOD   | 31878  |

| Reaction No. 54648                                                  |    |       |               |                             |   |       |        |
|---------------------------------------------------------------------|----|-------|---------------|-----------------------------|---|-------|--------|
| $\text{Ge(s)} + 4\text{H}^+ + 4\text{e}^- = \text{GeH}_4(\text{g})$ |    |       |               |                             |   |       |        |
| No                                                                  | t  | I-Str | Medium        | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0     | Inf. Dilution | $\lg K: -20.63(4\text{SF})$ | 1 | ELC   |        |
| 2                                                                   | 20 | 0     | Inf. Dilution | $E: -0.3(0.2\text{SD})$     | 2 | EES   | 775    |
| 3                                                                   | 25 | 0     | Inf. Dilution | $E: -0.42(2\text{SF})$      | 0 | CRV   | 22551  |

| Reaction No. 55165                                                              |    |       |         |                                |   |       |        |
|---------------------------------------------------------------------------------|----|-------|---------|--------------------------------|---|-------|--------|
| $\text{Ge}^{+4} + \text{F}^- = \text{Ge}^{+4}\text{F}^-$                        |    |       |         |                                |   |       |        |
| Note: $\text{Ge}^{+4}$ is deprecated because does not exist in aqueous solution |    |       |         |                                |   |       |        |
| No                                                                              | t  | I-Str | Medium  | Value (Dev)                    | W | Tech. | Ref. # |
| 1                                                                               | 20 | 0     | Unknown | $\lg K: 1.68(3\text{SF})$      | 0 | MSE   | 31904  |
| 2                                                                               | 25 | 0.4   | NaCl    | $\text{dH}: 2.58(5.\text{SD})$ | 0 | ENS   | 437    |

| Reaction No. 56784                                                     |    |       |               |                              |   |       |        |
|------------------------------------------------------------------------|----|-------|---------------|------------------------------|---|-------|--------|
| $\text{GeO}_2 + 2\text{OH}^- = \text{GeO}_2^{2-} + \text{H}_2\text{O}$ |    |       |               |                              |   |       |        |
| No                                                                     | t  | I-Str | Medium        | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                      | 25 | 0     | Inf. Dilution | $\lg K: 1.48(3\text{SF})$    | 1 | EDH   |        |
| 2                                                                      | 25 | 3     | NaCl          | $\lg K: 1.64(0.03\text{MD})$ | 6 | MHY   | 22980  |

| Reaction No. 57621                                                              |    |       |                    |                             |   |       |        |
|---------------------------------------------------------------------------------|----|-------|--------------------|-----------------------------|---|-------|--------|
| $\text{Ge}^{+4} + 4\text{OH}^- = \text{Ge}^{+4}\text{OH}_4^{4-}$                |    |       |                    |                             |   |       |        |
| Note: $\text{Ge}^{+4}$ is deprecated because does not exist in aqueous solution |    |       |                    |                             |   |       |        |
| No                                                                              | t  | I-Str | Medium             | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.1   | KNO <sub>3</sub>   | $\lg K: 55.6(3\text{SF})$   | 0 | MSP   | 31897  |
| 2                                                                               | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: 57.04(4\text{SF})$  | 0 | MGL   | 903    |
| 3                                                                               | 25 | 8     | LiCl               | $\lg K: 56.8(0.1\text{SD})$ | 0 | MSE   | 5339   |

| Reaction No. 57622                                                              |    |       |                  |                             |   |       |        |
|---------------------------------------------------------------------------------|----|-------|------------------|-----------------------------|---|-------|--------|
| $\text{Ge}^{+4} + 3\text{OH}^- = \text{Ge}^{+4}\text{OH}_3^{3-}$                |    |       |                  |                             |   |       |        |
| Note: $\text{Ge}^{+4}$ is deprecated because does not exist in aqueous solution |    |       |                  |                             |   |       |        |
| No                                                                              | t  | I-Str | Medium           | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.1   | KNO <sub>3</sub> | $\lg K: 42.2(3\text{SF})$   | 0 | MSP   | 31897  |
| 2                                                                               | 25 | 8     | LiCl             | $\lg K: 43.3(0.1\text{SD})$ | 0 | MSE   | 5339   |

| Reaction No. 57623 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| 2<OH-1> + Ge+4 = Ge+4_OH-1(2)                                       |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Note: Ge+4 is deprecated because does not exist in aqueous solution |    |       |        |                 |   |       |        |
| No                                                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.1   | KNO3   | lgK:28.50(4SF)  | 0 | MSP   | 31897  |
| 2                                                                   | 25 | 8     | LiCl   | lgK:29.2(0.1SD) | 0 | MSE   | 5339   |

| Reaction No. 57624                                                  |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ge+4 + OH-1 = Ge+4_OH-1                                             |    |       |        |                 |   |       |        |
| Note: Ge+4 is deprecated because does not exist in aqueous solution |    |       |        |                 |   |       |        |
| No                                                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.1   | KNO3   | lgK:14.45(4SF)  | 0 | MSP   | 31897  |
| 2                                                                   | 25 | 8     | LiCl   | lgK:14.8(0.1SD) | 0 | MSE   | 5339   |

| Reaction No. 57970                  |    |       |               |                       |   |       |        |
|-------------------------------------|----|-------|---------------|-----------------------|---|-------|--------|
| 2<F2(g)> + Ge(s) = Ge+4_F-1(4)_ (g) |    |       |               |                       |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)           | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | dG:-1150.0(0.584SD)kJ | 6 | CRV   | 14923  |
| 2                                   | 25 | 0     | Inf. Dilution | dH:-284.5(0.1SD)      | 7 | CRV   | 5605   |
| 3                                   | 25 | 0     | Inf. Dilution | dH:-1190.20(0.50MD)kJ | 6 | CRV   | 8746   |
| 4                                   | 25 | 0     | Inf. Dilution | dH:-1190.2(0.500SD)kJ | 6 | CRV   | 14923  |
| 5                                   | 25 | 0     | Inf. Dilution | Cp:81.602(0.5SD)J/K   | 5 | CRV   | 27292  |

| Reaction No. 58593                                                           |    |       |               |                  |   |       |        |
|------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1(2)_Citric-3> = Ge+4_H+1(2)_OH-1(2)_Citric-3(2) + 2<H2O> |    |       |               |                  |   |       |        |
| No                                                                           | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.1   | NaCl/m        | lgK:6.24(0.30SD) | 5 | MSL   | 8740   |
| 2                                                                            | 90 | 0.1   | NaCl/m        | lgK:4.68(0.10SD) | 5 | MSL   | 8740   |
| 3                                                                            | 25 | 0     | Inf. Dilution | dH:-11.65(4SF)   | 4 | MTD   | 8740   |

| Reaction No. 59433                                               |    |       |        |                   |   |       |        |
|------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Ge+4_OH-1(4) + 3<H+1(2)_Cat-2> = Ge+4_Cat-2(3) + 2<H+1> + 4<H2O> |    |       |        |                   |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                | 25 | 0.1   | NaCl/m | lgK:-1.39(0.05SD) | 5 | MSL   | 8740   |
| 2                                                                | 90 | 0.1   | NaCl/m | lgK:-4.05(0.10SD) | 5 | MSL   | 8740   |

|                                                                    |  |  |  |  |  |  |  |
|--------------------------------------------------------------------|--|--|--|--|--|--|--|
| Reaction No. 59434                                                 |  |  |  |  |  |  |  |
| Ge+4_OH-1(4) + 2<H+1_Oxalic-2> = Ge+4_OH-1(2)_Oxalic-2(2) + 2<H2O> |  |  |  |  |  |  |  |



| No                                                               | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 1                                                                | 25 | 0.2   | NaOH   | lgK:4.35 (3SF) | 4 | MSL   | 8740   |
| Note (1): Ionic media variable, either NaOH or mix NaCl and NaOH |    |       |        |                |   |       |        |

| Reaction No. 60204                                                  |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| EDTA-4 + Ge+4 = Ge+4_EDTA-4                                         |    |       |        |                 |   |       |        |
| Note: Ge+4 is deprecated because does not exist in aqueous solution |    |       |        |                 |   |       |        |
| No                                                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                   | 20 | 1     | NaClO4 | lgK:27.02 (4SF) | 0 | MGL   | 903    |

| Reaction No. 60228                            |    |       |         |                |   |       |        |
|-----------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| H+1 + Ge+4_OH-1_EDTA-4 = Ge+4_H+1_OH-1_EDTA-4 |    |       |         |                |   |       |        |
| No                                            | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 20 | 0.1   | KNO3    | lgK:2.64 (3SF) | 5 | MGL   | 3450   |
| 2                                             | 25 | 0.1   | Unknown | lgK:2.40 (3SF) | 0 | MGL   | 906    |
| 3                                             | 25 | 0.1   | Unknown | lgK:2.40 (3SF) | 4 | MPL   | 31978  |

| Reaction No. 60229                                                              |    |       |         |                |   |       |        |
|---------------------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Ge+4_OH-1(4) + H+1(4)_EDTA-4 = Ge+4_H+1_OH-1_EDTA-4 + 3<H2O>                    |    |       |         |                |   |       |        |
| Note: Complex initially Ge(OH)2H2L, later H[Ge(OH)L]; this form adopted in JESS |    |       |         |                |   |       |        |
| No                                                                              | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.1   | KClO4   | lgK:4.80 (3SF) | 3 | MPL   | 931    |
| Note (1): Value from 67Kon [31894], thus complex written as Ge(OH)2H2L          |    |       |         |                |   |       |        |
| 2                                                                               | 25 | 0.1   | KClO4   | lgK:4.80 (3SF) | 5 | MPL   | 31894  |
| Note (2): Complex written Ge(OH)2H2L; modified later by the authors             |    |       |         |                |   |       |        |
| 3                                                                               | 25 | 0.1   | KClO4   | lgK:4.80 (3SF) | 5 | MGL   | 31978  |
| 4                                                                               | 25 | 0.1   | NaClO4  | lgK:4.80 (3SF) | 3 | MGL   | 903    |
| 5                                                                               | 25 | 0.1   | Unknown | lgK:4.80 (3SF) | 4 | MGL   | 906    |
| 6                                                                               | 25 | 0.5   | NaCl    | lgK:5.3 (2SF)  | 2 | MPT   | 906    |

| Reaction No. 60230                                                                    |    |       |        |                    |   |       |        |
|---------------------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 2<Glucosaminic-1> + Ge+4_OH-1(4) + H+1 = Ge+4_H+1(-2)_Glucosaminic-1(2)_OH-1 + 3<H2O> |    |       |        |                    |   |       |        |
| No                                                                                    | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                     | 25 | 0.1   | NaClO4 | lgK:14.35 (0.04SD) | 5 | MGL   | 9211   |

|                    |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
| Reaction No. 60231 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|

| Glucosaminic-1 + Ge+4_OH-1(4) = GeO+2_H+1(-2)_Glucosaminic-1_OH-1 + 2<H2O> + H+1 |    |       |        |                |   |       |        |
|----------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                                                                               | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                | 25 | 0.1   | NaClO4 | lgK:-6.23(3SF) | 5 | MGL   | 9211   |

| Reaction No. 60232                                                  |    |       |         |              |   |       |        |
|---------------------------------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| Ge+4 + H+1(4)_EDTA-4 = Ge+4_EDTA-4 + 4<H+1>                         |    |       |         |              |   |       |        |
| Note: Ge+4 is deprecated because does not exist in aqueous solution |    |       |         |              |   |       |        |
| No                                                                  | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                   | 24 | 0.5   | Unknown | lgK:5.3(2SF) | 0 | MGL   | 903    |

| Reaction No. 60546                  |    |       |               |                 |   |       |        |
|-------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| GeO(black,s) + H2O = Ge+2 + 2<OH-1> |    |       |               |                 |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:-32.58(4SF) | 1 | ELC   |        |
| 2                                   | 25 | 0     | Inf. Dilution | lgK:-3.7(2SF)   | 0 | CRV   | 2556   |

| Reaction No. 60706                                                    |    |       |               |                  |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Ge+4_OH-1(4) + 4<H+1_F-1> = Ge+4_F-1(4) + 4<H2O>                      |    |       |               |                  |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0     | Inf. Dilution | lgK:7.6(2SF)     | 1 | EDH   |        |
| 2                                                                     | 25 | 0.5   | LiClO4        | lgK:7.5(0.1SD)   | 3 | MPT   | 31881  |
| Note (2): Rec. in [9906], considering ML4 and ML6 instead ML4 and ML5 |    |       |               |                  |   |       |        |
| 3                                                                     | 25 | 0.5   | Unknown       | lgK:7.5(2SF)     | 2 | CRV   | 2556   |
| 4                                                                     | 25 | 2.5   | LiClO4        | lgK:7.7(2SF)     | 0 | CRV   | 2556   |
| 5                                                                     | 25 | 2.5   | NaClO4        | lgK:7.2(2SF)     | 5 | CRV   | 2556   |
| 6                                                                     | 25 | 3     | LiClO4        | lgK:7.73(0.02SD) | 0 | MGL   | 9906   |
| 7                                                                     | 25 | 3     | NaClO4        | lgK:7.18(0.10SD) | 5 | MGL   | 9906   |

| Reaction No. 60948                                                  |    |       |         |               |   |       |        |
|---------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Ge+4 + 5<F-1> = Ge+4_F-1(5)                                         |    |       |         |               |   |       |        |
| Note: Ge+4 is deprecated because does not exist in aqueous solution |    |       |         |               |   |       |        |
| No                                                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                   | 20 | 0     | Unknown | lgK:6.07(3SF) | 0 | MSE   | 31904  |

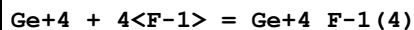
| Reaction No. 64286                    |   |       |        |             |   |       |        |
|---------------------------------------|---|-------|--------|-------------|---|-------|--------|
| GeO2(hexag,s) + 2<H2O> = Ge+4_OH-1(4) |   |       |        |             |   |       |        |
| No                                    | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|                                            |     |   |             |                 |   |     |       |
|--------------------------------------------|-----|---|-------------|-----------------|---|-----|-------|
| 1                                          | 0   | 0 | Self Medium | lgK:-1.631(4SF) | 3 | EOD | 31908 |
| Note (1): Source data: Vehov et al (1964)  |     |   |             |                 |   |     |       |
| 2                                          | 0   | 0 | Self Medium | lgK:-1.7(2SF)   | 1 | MSL | 31960 |
| 3                                          | 11  | 0 | Self Medium | lgK:-1.42(3SF)  | 2 | MSL | 31869 |
| 4                                          | 18  | 0 | Self Medium | lgK:-1.4(2SF)   | 1 | MSL | 31960 |
| 5                                          | 20  | 0 | Self Medium | lgK:-1.39(3SF)  | 2 | MSL | 31869 |
| 6                                          | 20  | 0 | Self Medium | lgK:-1.4(2SF)   | 1 | MSL | 32019 |
| 7                                          | 25  | 0 | Self Medium | lgK:-1.36(3SF)  | 2 | MSL | 7864  |
| 8                                          | 25  | 0 | Self Medium | lgK:-1.38(3SF)  | 5 | MSL | 8740  |
| 9                                          | 25  | 0 | Self Medium | lgK:-1.37(3SF)  | 3 | MSL | 31868 |
| 10                                         | 25  | 0 | Self Medium | lgK:-1.370(4SF) | 4 | EOD | 31908 |
| Note (10): Source data: Vehov et al (1964) |     |   |             |                 |   |     |       |
| 11                                         | 25  | 0 | Self Medium | lgK:-1.37(3SF)  | 0 | MSL | 31959 |
| 12                                         | 25  | 0 | Self Medium | lgK:-1.38(3SF)  | 2 | MSL | 31971 |
| 13                                         | 26  | 0 | Self Medium | lgK:-1.35(3SF)  | 2 | MSL | 31869 |
| 14                                         | 30  | 0 | Self Medium | lgK:-1.4(2SF)   | 2 | MSL | 32018 |
| 15                                         | 35  | 0 | Self Medium | lgK:-1.28(3SF)  | 2 | MSL | 31869 |
| 16                                         | 41  | 0 | Self Medium | lgK:-1.23(3SF)  | 2 | MSL | 31869 |
| 17                                         | 50  | 0 | Self Medium | lgK:-1.168(4SF) | 4 | EOD | 31908 |
| Note (17): Source data: Vehov et al (1964) |     |   |             |                 |   |     |       |
| 18                                         | 75  | 0 | Self Medium | lgK:-0.994(3SF) | 2 | EOD | 31908 |
| Note (18): Source data: Vehov et al (1964) |     |   |             |                 |   |     |       |
| 19                                         | 100 | 0 | Self Medium | lgK:-0.848(3SF) | 2 | EOD | 31908 |
| Note (19): Source data: Vehov et al (1964) |     |   |             |                 |   |     |       |
| 20                                         | 100 | 0 | Self Medium | lgK:-1.0(2SF)   | 1 | MSL | 32019 |

| Reaction No. 65966                                                    |    |       |               |                 |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 6<H+1_F-1> = Ge+4_H+1_F-1(6) + H+1 + 4<H2O>            |    |       |               |                 |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0     | Inf. Dilution | lgK:9.9(2SF)    | 1 | EDH   |        |
| 2                                                                     | 25 | 0.5   | LiClO4        | lgK:10.2(0.1SD) | 2 | MPT   | 31881  |
| Note (2): Rec. in [9906], considering ML4 and ML6 instead ML4 and ML5 |    |       |               |                 |   |       |        |
| 3                                                                     | 25 | 0.5   | Unknown       | lgK:10.2(3SF)   | 2 | CRV   | 552    |
| 4                                                                     | 25 | 2.5   | LiClO4        | lgK:10.8(3SF)   | 0 | CRV   | 552    |
| 5                                                                     | 25 | 2.5   | NaClO4        | lgK:9.9(2SF)    | 2 | CRV   | 552    |
| 6                                                                     | 25 | 3     | LiClO4        | lgK:10.8(0.1SD) | 0 | MGL   | 9906   |

|   |    |   |                    |                   |   |     |      |
|---|----|---|--------------------|-------------------|---|-----|------|
| 7 | 25 | 3 | NaClO <sub>4</sub> | lgK:9.94 (0.10SD) | 4 | MGL | 9906 |
|---|----|---|--------------------|-------------------|---|-----|------|

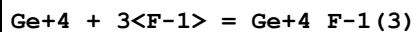
## Reaction No. 66135



Note: Ge+4 is deprecated because does not exist in aqueous solution

| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0     | Unknown | lgK:5.17 (3SF) | 0 | MSE   | 31904  |

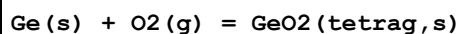
## Reaction No. 66136



Note: Ge+4 is deprecated because does not exist in aqueous solution

| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0     | Unknown | lgK:4.18 (3SF) | 0 | MSE   | 31904  |

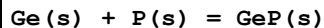
## Reaction No. 66760



| No | t   | I-Str | Medium        | Value (Dev)             | W | Tech. | Ref. # |
|----|-----|-------|---------------|-------------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:85. (2SF)           | 0 | ELC   |        |
| 2  | 25  | 0     | Inf. Dilution | lgK:91.33 (4SF)         | 0 | ENS   | 6422   |
| 3  | 27  | 0     | Inf. Dilution | lgK:90.70 (4SF)         | 0 | ENS   | 6422   |
| 4  | 127 | 0     | Inf. Dilution | lgK:65.46 (4SF)         | 4 | ENS   | 6422   |
| 5  | 227 | 0     | Inf. Dilution | lgK:50.33 (4SF)         | 3 | ENS   | 6422   |
| 6  | 327 | 0     | Inf. Dilution | lgK:40.26 (4SF)         | 0 | ENS   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dG:-124.6 (4SF)         | 0 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dG:-490. (3SF) kJ       | 0 | CRV   | 8746   |
| 9  | 25  | 0     | Inf. Dilution | dG:-521.40 (1.002SD) kJ | 0 | CRV   | 14923  |
| 10 | 25  | 0     | Inf. Dilution | dG:-521.36 (2.93SD) kJ  | 0 | EOD   | 30883  |
| 11 | 25  | 0     | Inf. Dilution | dG:-124.60 (5SF)        | 0 | CRV   | 31291  |
| 12 | 25  | 0     | Inf. Dilution | dH:-139. (0.3SD)        | 7 | CRV   | 5605   |
| 13 | 25  | 0     | Inf. Dilution | dH:-138.6 (4SF)         | 4 | MTD   | 6422   |
| 14 | 25  | 0     | Inf. Dilution | dH:-580.0 (1.0MD) kJ    | 6 | CRV   | 8746   |
| 15 | 25  | 0     | Inf. Dilution | dH:-580.00 (1.000SD) kJ | 6 | CRV   | 14923  |
| 16 | 25  | 0     | Inf. Dilution | dH:-580.32 (2.93SD) kJ  | 3 | EOD   | 30883  |
| 17 | 25  | 0     | Inf. Dilution | dH:-138.60 (5SF)        | 3 | CRV   | 31291  |
| 18 | 127 | 0     | Inf. Dilution | dH:-138.5 (4SF)         | 2 | MTD   | 6422   |
| 19 | 227 | 0     | Inf. Dilution | dH:-138.3 (4SF)         | 2 | MTD   | 6422   |

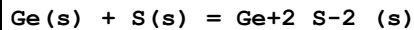
|    |     |   |               |                      |   |     |       |
|----|-----|---|---------------|----------------------|---|-----|-------|
| 20 | 327 | 0 | Inf. Dilution | dH:-138.1(4SF)       | 0 | MTD | 6422  |
| 21 | 25  | 0 | Inf. Dilution | Cp:50.166(0.3SD) J/K | 5 | CRV | 27292 |
| 22 | 25  | 0 | Inf. Dilution | Cp:68.86(4SF) J/K    | 3 | EOD | 30883 |

## Reaction No. 66761



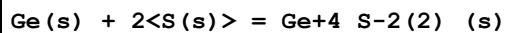
| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:3.201(4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:3.179(4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:2.228(4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:1.636(4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:1.235(4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-4.37(3SF)  | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-5.0(2SF)   | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-5.3(2SF)   | 4 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-5.5(2SF)   | 4 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-5.6(2SF)   | 4 | MTD   | 6422   |

## Reaction No. 66762



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:13.49(4SF)   | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:13.41(4SF)   | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:10.08(4SF)   | 2 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:8.026(4SF)   | 1 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:6.634(4SF)   | 0 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-18.40(4SF)   | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dG:-71.5(3SF) kJ | 3 | CRV   | 9015   |
| 8  | 25  | 0     | Inf. Dilution | dH:-18.2(3SF)    | 4 | MTD   | 6422   |
| 9  | 25  | 0     | Inf. Dilution | dH:-69.0(3SF) kJ | 4 | CRV   | 9015   |
| 10 | 127 | 0     | Inf. Dilution | dH:-18.7(3SF)    | 1 | MTD   | 6422   |
| 11 | 227 | 0     | Inf. Dilution | dH:-19.0(3SF)    | 0 | MTD   | 6422   |
| 12 | 327 | 0     | Inf. Dilution | dH:-19.2(3SF)    | 0 | MTD   | 6422   |

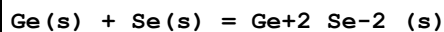
## Reaction No. 66763



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

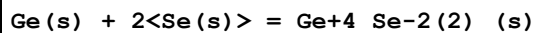
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 1  | 25  | 0 | Inf. Dilution | lgK:27.08 (4SF) | 4 | ENS | 6422 |
| 2  | 27  | 0 | Inf. Dilution | lgK:26.91 (4SF) | 4 | ENS | 6422 |
| 3  | 127 | 0 | Inf. Dilution | lgK:20.05 (4SF) | 4 | ENS | 6422 |
| 4  | 227 | 0 | Inf. Dilution | lgK:15.79 (4SF) | 4 | ENS | 6422 |
| 5  | 327 | 0 | Inf. Dilution | lgK:12.90 (4SF) | 4 | ENS | 6422 |
| 6  | 25  | 0 | Inf. Dilution | dG:-36.95 (4SF) | 4 | EFE | 6422 |
| 7  | 25  | 0 | Inf. Dilution | dH:-37.5 (3SF)  | 4 | MTD | 6422 |
| 8  | 127 | 0 | Inf. Dilution | dH:-38.7 (3SF)  | 4 | MTD | 6422 |
| 9  | 227 | 0 | Inf. Dilution | dH:-39.4 (3SF)  | 4 | MTD | 6422 |
| 10 | 327 | 0 | Inf. Dilution | dH:-40.0 (3SF)  | 4 | MTD | 6422 |

## Reaction No. 66764



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:12.35 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:12.28 (4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:9.272 (4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:7.459 (4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:6.147 (4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-16.85 (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-16.5 (3SF)  | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-16.5 (3SF)  | 4 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-17.9 (3SF)  | 4 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-18.1 (3SF)  | 4 | MTD   | 6422   |

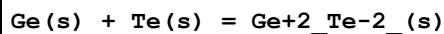
## Reaction No. 66765



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:19.63 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:19.51 (4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:14.58 (4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:11.60 (4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:9.387 (4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-26.78 (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-27.0 (3SF)  | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-27.1 (3SF)  | 4 | MTD   | 6422   |

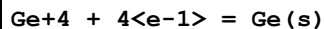
|    |     |   |               |               |   |     |      |
|----|-----|---|---------------|---------------|---|-----|------|
| 9  | 227 | 0 | Inf. Dilution | dH:-30.1(3SF) | 4 | MTD | 6422 |
| 10 | 327 | 0 | Inf. Dilution | dH:-30.5(3SF) | 4 | MTD | 6422 |

## Reaction No. 66766



| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:8.993(4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:8.940(4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:6.832(4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:5.569(4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:4.726(4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-12.27(4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-11.6(3SF)  | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-11.6(3SF)  | 4 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-11.6(3SF)  | 4 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-11.6(3SF)  | 4 | MTD   | 6422   |

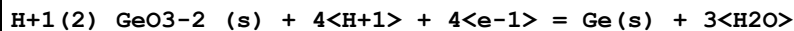
## Reaction No. 71062



Note: Ge<sub>4</sub> is deprecated because does not exist in aqueous solution

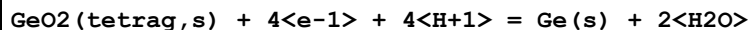
| No | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:8.1(2SF) | 0 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | E:0.124(3SF) | 0 | CRV   | 6330   |
| 3  | 25 | 0     | Inf. Dilution | E:0.124(3SF) | 0 | CRV   | 22551  |

## Reaction No. 71063



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | E:-0.182(3SF) | 5 | CRV   | 6330   |

## Reaction No. 74241



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-3.86(3SF)  | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | E:-0.05(0.01SD) | 2 | MDG   | 775    |
| 3  | 25 | 0     | Inf. Dilution | E:-0.058(2SF)   | 0 | CRV   | 22551  |
| 4  | 25 | 0     | Inf. Dilution | E:-0.053(2SF)   | 3 | EOD   | 31878  |

| Reaction No. 74242                                                                                                                               |    |       |               |                           |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------------------|---|-------|--------|
| $\text{Ge}+4_{\text{OH}-1}(4) + 4\langle\text{e}-1\rangle + 4\langle\text{H}+1\rangle = \text{Ge}(\text{s}) + 4\langle\text{H}_2\text{O}\rangle$ |    |       |               |                           |   |       |        |
| No                                                                                                                                               | t  | I-Str | Medium        | Value (Dev)               | W | Tech. | Ref. # |
| 1                                                                                                                                                | 25 | 0     | Inf. Dilution | $\lg K: 0.71(2\text{SF})$ | 1 | ELC   |        |
| 2                                                                                                                                                | 25 | 0     | Inf. Dilution | $E: 0.01(1\text{SF})$     | 3 | EOD   | 775    |
| 3                                                                                                                                                | 25 | 0     | Inf. Dilution | $E: 0.012(2\text{SF})$    | 0 | CRV   | 22551  |
| 4                                                                                                                                                | 25 | 0     | Inf. Dilution | $E: 0.011(2\text{SF})$    | 3 | EOD   | 31878  |

| Reaction No. 74243                                                                                                                                             |    |       |               |                           |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------------------|---|-------|--------|
| $\text{Ge}+4_{\text{OH}-1}(4) + 2\langle\text{e}-1\rangle + 2\langle\text{H}+1\rangle = \text{GeO}(\text{black},\text{s}) + 3\langle\text{H}_2\text{O}\rangle$ |    |       |               |                           |   |       |        |
| No                                                                                                                                                             | t  | I-Str | Medium        | Value (Dev)               | W | Tech. | Ref. # |
| 1                                                                                                                                                              | 25 | 0     | Inf. Dilution | $\lg K: -2.8(2\text{SF})$ | 1 | ELC   |        |
| 2                                                                                                                                                              | 25 | 0     | Inf. Dilution | $E: -0.72(2\text{SF})$    | 0 | CRV   | 22551  |
| Note (2): $\text{H}+1(2)_{\text{GeO}_3(2)}$ in source                                                                                                          |    |       |               |                           |   |       |        |

| Reaction No. 74244                                                                                                                       |    |       |               |                            |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------------------|---|-------|--------|
| $\text{Ge}+4_{\text{OH}-1}(4) + 2\langle\text{e}-1\rangle + 4\langle\text{H}+1\rangle = \text{Ge}+2 + 4\langle\text{H}_2\text{O}\rangle$ |    |       |               |                            |   |       |        |
| No                                                                                                                                       | t  | I-Str | Medium        | Value (Dev)                | W | Tech. | Ref. # |
| 1                                                                                                                                        | 25 | 0     | Inf. Dilution | $\lg K: -7.19(3\text{SF})$ | 1 | ELC   |        |
| 2                                                                                                                                        | 25 | 0     | Inf. Dilution | $E: -0.223(3\text{SF})$    | 0 | CRV   | 22551  |
| Note (2): $\text{H}+1(2)_{\text{GeO}_3(2)}$ in source                                                                                    |    |       |               |                            |   |       |        |

| Reaction No. 74245                                                                                                                     |    |       |               |                            |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------------------|---|-------|--------|
| $\text{GeO}(\text{black},\text{s}) + 2\langle\text{e}-1\rangle + 2\langle\text{H}+1\rangle = \text{Ge}(\text{s}) + \text{H}_2\text{O}$ |    |       |               |                            |   |       |        |
| No                                                                                                                                     | t  | I-Str | Medium        | Value (Dev)                | W | Tech. | Ref. # |
| 1                                                                                                                                      | 25 | 0     | Inf. Dilution | $\lg K: -2.99(3\text{SF})$ | 0 | ELC   |        |
| 2                                                                                                                                      | 25 | 0     | Inf. Dilution | $E: 0.095(2\text{SF})$     | 0 | CRV   | 22551  |
| 3                                                                                                                                      | 25 | 0     | Inf. Dilution | $E: 0.100(3\text{SF})$     | 3 | EOD   | 31878  |

| Reaction No. 74246                                                                                                     |    |       |               |                             |   |       |        |
|------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------------------|---|-------|--------|
| $\text{GeO}_2(\text{hexag},\text{s}) + 2\langle\text{H}_2\text{O}\rangle = \text{H}+1 + \text{GeO}+2_{\text{OH}-1}(3)$ |    |       |               |                             |   |       |        |
| No                                                                                                                     | t  | I-Str | Medium        | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                                                                      | 25 | 0     | Inf. Dilution | $\lg K: -10.40(4\text{SF})$ | 1 | ELC   |        |
| 2                                                                                                                      | 25 | 0     | Inf. Dilution | $\lg K: -9.96(3\text{SF})$  | 0 | CRV   | 22551  |
| Note (2): Phase not specified in source / $\text{H}+1_{\text{GeO}_3-2}$ in source                                      |    |       |               |                             |   |       |        |



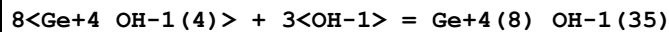
| Reaction No. 75138                                                  |    |       |         |               |   |       |        |
|---------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| $\text{Ge}+4 + 2\text{<F-1>} = \text{Ge}+4\_F-1(2)$                 |    |       |         |               |   |       |        |
| Note: Ge+4 is deprecated because does not exist in aqueous solution |    |       |         |               |   |       |        |
| No                                                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                   | 20 | 0     | Unknown | lgK:3.03(3SF) | 0 | MSE   | 31904  |

| Reaction No. 75139                                                         |     |       |               |                    |   |       |        |
|----------------------------------------------------------------------------|-----|-------|---------------|--------------------|---|-------|--------|
| $\text{Ge}+4\_OH-1(4) + OH-1 = \text{GeO}+2\_OH-1(3) + \text{H}_2\text{O}$ |     |       |               |                    |   |       |        |
| No                                                                         | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                          | 25  | 0     | Inf. Dilution | lgK:4.4(2SF)       | 1 | ELC   |        |
| 2                                                                          | 25  | 0     | Self Medium   | lgK:4.68(3SF)      | 4 | MSL   | 31910  |
| Note (2): Calculated from equation 10 in paper                             |     |       |               |                    |   |       |        |
| 3                                                                          | 25  | 0.1   | NaCl          | lgK:4.68(3SF)      | 2 | MGL   | 140    |
| 4                                                                          | 25  | 0.5   | Na(Cl)        | lgK:4.678(0.006MD) | 6 | MHY   | 22981  |
| 5                                                                          | 25  | 0.5   | NaClO4        | lgK:4.704(0.002SD) | 5 | MGL   | 31887  |
| 6                                                                          | 25  | 1     | NaClO4        | lgK:4.763(0.07SD)  | 5 | MGL   | 31887  |
| 7                                                                          | 50  | 0     | Self Medium   | lgK:3.93(3SF)      | 4 | MSL   | 31910  |
| Note (7): Calculated from equation 10 in paper                             |     |       |               |                    |   |       |        |
| 8                                                                          | 100 | 0     | Self Medium   | lgK:2.77(3SF)      | 4 | MSL   | 31910  |
| Note (8): Calculated from equation 10 in paper                             |     |       |               |                    |   |       |        |
| 9                                                                          | 150 | 0     | Self Medium   | lgK:1.94(3SF)      | 4 | MSL   | 31910  |
| Note (9): Calculated from equation 10 in paper                             |     |       |               |                    |   |       |        |
| 10                                                                         | 200 | 0     | Self Medium   | lgK:1.32(3SF)      | 4 | MSL   | 31910  |
| Note (10): Calculated from equation 10 in paper                            |     |       |               |                    |   |       |        |
| 11                                                                         | 250 | 0     | Self Medium   | lgK:0.873(3SF)     | 4 | MSL   | 31910  |
| Note (11): Calculated from equation 10 in paper                            |     |       |               |                    |   |       |        |
| 12                                                                         | 300 | 0     | Self Medium   | lgK:0.536(3SF)     | 4 | MSL   | 31910  |
| Note (12): Calculated from equation 10 in paper                            |     |       |               |                    |   |       |        |

| Reaction No. 75140                                                                      |    |       |               |                  |   |       |        |
|-----------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{Ge}+4\_OH-1(4) + 2\text{<OH-1>} = \text{GeO}+2\_OH-1(2) + 2\text{<H}_2\text{O>}$ |    |       |               |                  |   |       |        |
| No                                                                                      | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                       | 25 | 0     | Inf. Dilution | lgK:4.8(2SF)     | 0 | EDH   |        |
| 2                                                                                       | 25 | 0     | Inf. Dilution | lgK:6.15(3SF)    | 1 | ELC   |        |
| 3                                                                                       | 25 | 0.5   | Na(Cl)        | lgK:6.14(0.13MD) | 0 | MHY   | 22981  |
| 4                                                                                       | 25 | 0.5   | NaClO4        | lgK:5.60(0.48SD) | 0 | MGL   | 31887  |

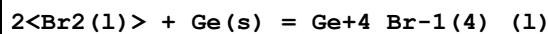
|   |    |   |                    |                   |   |     |       |
|---|----|---|--------------------|-------------------|---|-----|-------|
| 5 | 25 | 1 | NaClO <sub>4</sub> | lgK:6.80 (0.07SD) | 3 | MGL | 31887 |
|---|----|---|--------------------|-------------------|---|-----|-------|

## Reaction No. 75141



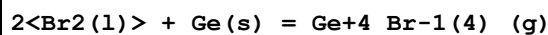
| No | t  | I-Str | Medium             | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution      | lgK:28.4 (3SF)     | 1 | EDH   |        |
| 2  | 25 | 0.5   | Na (Cl)            | lgK:29.14 (0.05MD) | 4 | MHY   | 22981  |
| 3  | 25 | 0.5   | NaClO <sub>4</sub> | lgK:29.55 (0.01SD) | 2 | MGL   | 31887  |
| 4  | 25 | 1     | NaClO <sub>4</sub> | lgK:30.37 (0.02SD) | 0 | MGL   | 31887  |

## Reaction No. 75593



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-331.4 (4SF) kJ | 3 | CRV   | 6330   |
| 2  | 25 | 0     | Inf. Dilution | dH:-347.7 (4SF) kJ | 3 | CRV   | 6330   |

## Reaction No. 75594



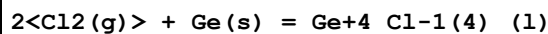
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-318.0 (4SF) kJ | 3 | CRV   | 6330   |
| 2  | 25 | 0     | Inf. Dilution | dH:-300.0 (4SF) kJ | 3 | CRV   | 6330   |

## Reaction No. 75600



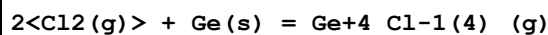
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:124.2 (4SF) kJ | 3 | CRV   | 6330   |
| 2  | 25 | 0     | Inf. Dilution | dH:155.2 (4SF) kJ | 3 | CRV   | 6330   |

## Reaction No. 75622



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-462.7 (4SF) kJ | 3 | CRV   | 6330   |
| 2  | 25 | 0     | Inf. Dilution | dH:-531.8 (4SF) kJ | 3 | CRV   | 6330   |

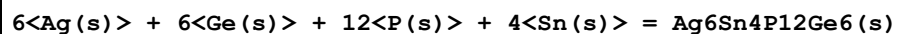
## Reaction No. 75623



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

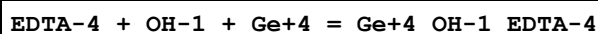
|   |    |   |               |                    |   |     |      |
|---|----|---|---------------|--------------------|---|-----|------|
| 1 | 25 | 0 | Inf. Dilution | dG:-457.3 (4SF) kJ | 3 | CRV | 6330 |
| 2 | 25 | 0 | Inf. Dilution | dH:-495.8 (4SF) kJ | 3 | CRV | 6330 |

## Reaction No. 76751



| No | t  | I-Str | Medium        | Value (Dev)            | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-76.485 (4.SD) kJ   | 5 | CRV   | 27323  |
| 2  | 25 | 0     | Inf. Dilution | dH:-147.000 (3.5SD) kJ | 5 | CRV   | 27323  |
| 3  | 25 | 0     | Inf. Dilution | Cp:628.820 (4.7SD) J/K | 5 | CRV   | 27323  |

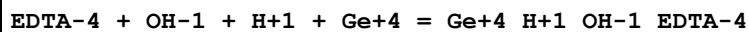
## Reaction No. 78252



Note: Ge+4 is deprecated because it does not exist in aqueous solution

| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:42.47 (4SF) | 0 | ELC   |        |

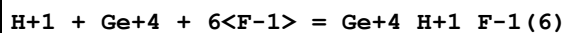
## Reaction No. 78639



Note: Ge+4 is deprecated because does not exist in aqueous solution

| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:44.88 (4SF) | 0 | ELC   |        |

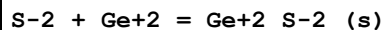
## Reaction No. 78896



Note: Ge+4 is deprecated because it does not exist in aqueous solution

| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:31.79 (4SF) | 0 | ELC   |        |

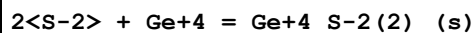
## Reaction No. 78971



Note: The species S-2 does not exist in aqueous solution See reaction:  $\text{H+1} + \text{S-2} = \text{H+1\_S-2}$

| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:36.59 (4SF) | 0 | ELC   |        |

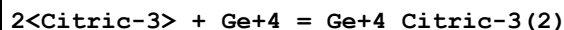
## Reaction No. 78972



Note: The species S-2 does not exist in aqueous solution See reaction:  $H+1 + S-2 = H+1\_S-2$

| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:65.51(4SF) | 0 | ELC   |        |

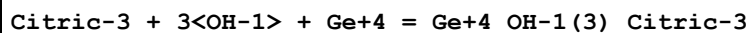
## Reaction No. 79197



Note: Ge+4 is deprecated because does not exist in aqueous solution

| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:39.6(3SF) | 0 | ELC   |        |

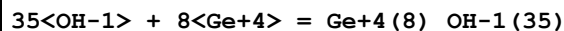
## Reaction No. 79198



Note: Ge+4 is deprecated because does not exist in aqueous solution

| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:62.0(3SF) | 0 | ELC   |        |

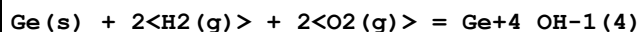
## Reaction No. 79385



Note: Ge+4 is deprecated because does not exist in aqueous solution

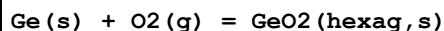
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:501.2(4SF) | 0 | ELC   |        |

## Reaction No. 80407



| No                                                                   | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 1                                                                    | 25 | 0     | Inf. Dilution | lgK:165.5(4SF)     | 1 | ELC   |        |
| 2                                                                    | 25 | 0     | Inf. Dilution | dG:-896.6(1.5SD)kJ | 0 | CRV   | 30883  |
| Note (2): Corresponding value for Si(OH) <sub>4</sub> is problematic |    |       |               |                    |   |       |        |
| 3                                                                    | 25 | 0     | Inf. Dilution | dG:-231.1(1.0SD)   | 0 | MSL   | 30959  |
| 4                                                                    | 25 | 0     | Inf. Dilution | dH:-265.7(1.5SD)   | 2 | MSL   | 30959  |
| 5                                                                    | 25 | 0     | Inf. Dilution | Cp:45.4(2.0SD)     | 2 | MSL   | 30959  |

## Reaction No. 80419



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:83.75(4SF) | 1 | ELC   |        |

|   |    |   |               |                         |   |     |       |
|---|----|---|---------------|-------------------------|---|-----|-------|
| 2 | 25 | 0 | Inf. Dilution | dG: -500.82 (1.25SD) kJ | 0 | EOD | 30883 |
| 3 | 25 | 0 | Inf. Dilution | dG: -119.69 (5SF)       | 0 | CRV | 31291 |
| 4 | 25 | 0 | Inf. Dilution | dH: -554.80 (1.25SD) kJ | 0 | EOD | 30883 |
| 5 | 25 | 0 | Inf. Dilution | dH: -132.58 (5SF)       | 0 | CRV | 31291 |
| 6 | 25 | 0 | Inf. Dilution | Cp: 61.18 (4SF) J/K     | 0 | EOD | 30883 |

| Reaction No. 80892                                          |    |       |        |                  |   |       |        |
|-------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1_F-1> + H+1 = Ge+4_F-1(2)_OH-1 + 3<H2O> |    |       |        |                  |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                           | 25 | 3     | LiClO4 | lgK: 3.2 (0.2SD) | 4 | MGL   | 9906   |
| 2                                                           | 25 | 3     | NaClO4 | lgK: 2.9 (0.1SD) | 4 | MGL   | 9906   |

| Reaction No. 80893                                          |    |       |        |                    |   |       |        |
|-------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Ge+4_OH-1(4) + 4<H+1_F-1> = Ge+4_F-1(4)_OH-1 + 3<H2O> + H+1 |    |       |        |                    |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                           | 25 | 3     | NaClO4 | lgK: 6.65 (0.05SD) | 5 | MGL   | 9906   |

| Reaction No. 80894                                        |    |       |        |                    |   |       |        |
|-----------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Ge+4_OH-1(4) + 6<H+1_F-1> = Ge+4_F-1(6) + 4<H2O> + 2<H+1> |    |       |        |                    |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                         | 25 | 3     | NaClO4 | lgK: 9.59 (0.05SD) | 5 | MGL   | 9906   |

| Reaction No. 80895                                  |    |       |             |                     |   |       |        |
|-----------------------------------------------------|----|-------|-------------|---------------------|---|-------|--------|
| Ge+4_OH-1(4) + H+1 + Cl-1 = Ge+4_Cl-1_OH-1(3) + H2O |    |       |             |                     |   |       |        |
| No                                                  | t  | I-Str | Medium      | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Self Medium | lgK: -3.02 (0.29SD) | 4 | MSE   | 31912  |

| Reaction No. 80896                                          |    |       |         |                     |   |       |        |
|-------------------------------------------------------------|----|-------|---------|---------------------|---|-------|--------|
| Ge+4_Cl-1_OH-1(3) + H+1 + Cl-1 = Ge+4_Cl-1(2)_OH-1(2) + H2O |    |       |         |                     |   |       |        |
| No                                                          | t  | I-Str | Medium  | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                           | 25 | 0     | Unknown | lgK: -3.84 (0.31SD) | 4 | MSE   | 31912  |

| Reaction No. 80897                                          |    |       |         |                     |   |       |        |
|-------------------------------------------------------------|----|-------|---------|---------------------|---|-------|--------|
| Ge+4_Cl-1(2)_OH-1(2) + H+1 + Cl-1 = Ge+4_Cl-1(3)_OH-1 + H2O |    |       |         |                     |   |       |        |
| No                                                          | t  | I-Str | Medium  | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                           | 25 | 0     | Unknown | lgK: -4.82 (0.51SD) | 4 | MSE   | 31912  |

| Reaction No. 80898 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Ge+4_C1-1(3)_OH-1 + H+1 + Cl-1 = Ge+4_C1-1(4) + H2O |    |       |         |                   |   |       |        |
|-----------------------------------------------------|----|-------|---------|-------------------|---|-------|--------|
| No                                                  | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Unknown | lgK:-5.09(0.56SD) | 4 | MSE   | 31912  |

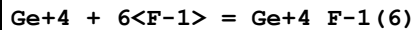
| Reaction No. 80899                                       |    |       |        |                     |   |       |        |
|----------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| Ge+4_OH-1(4) + DMannitol = GeO+2_DMannitol_OH-1(3) + H+1 |    |       |        |                     |   |       |        |
| No                                                       | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                        | 25 | 0.5   | NaCl   | lgK:-6.428(0.037SD) | 5 | MGL   | 7778   |

| Reaction No. 80900                                                       |    |       |        |                    |   |       |        |
|--------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 2<Ge+4_OH-1(4)> + 2<DMannitol> = GeO+2(2)_DMannitol(2)_OH-1(6) + 2<H+1 > |    |       |        |                    |   |       |        |
| No                                                                       | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                        | 25 | 0.5   | NaCl   | lgK:-10.62(0.08SD) | 5 | MGL   | 7778   |

| Reaction No. 80901                     |     |       |               |                   |   |       |        |
|----------------------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| GeO2(tetrag,s) + 2<H2O> = Ge+4_OH-1(4) |     |       |               |                   |   |       |        |
| No                                     | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                      | 25  | 0     | Inf. Dilution | lgK:-4.37(3SF)    | 3 | CRV   | 403    |
| 2                                      | 25  | 0     | Self Medium   | lgK:-5.02(0.13SD) | 5 | MSL   | 8763   |
| 3                                      | 25  | 0     | Self Medium   | lgK:-5.34(0.3SD)  | 5 | MSL   | 31908  |
| 4                                      | 25  | 0     | Self Medium   | lgK:-4.23(0.13SD) | 2 | EFE   | 31910  |
| 5                                      | 25  | 0     | Self Medium   | lgK:-4.37(3SF)    | 2 | MSL   | 31959  |
| 6                                      | 50  | 0     | Self Medium   | lgK:-4.57(0.20SD) | 5 | MSL   | 8763   |
| 7                                      | 50  | 0     | Self Medium   | lgK:-4.54(0.2SD)  | 4 | MSL   | 31908  |
| 8                                      | 90  | 0     | Self Medium   | lgK:-3.96(0.15SD) | 5 | MSL   | 8763   |
| 9                                      | 100 | 0     | Self Medium   | lgK:-3.42(0.15SD) | 4 | MSL   | 31908  |
| 10                                     | 100 | 0     | Self Medium   | lgK:-3.35(3SF)    | 2 | EFE   | 31910  |
| 11                                     | 150 | 0     | Self Medium   | lgK:-3.26(0.12SD) | 5 | MSL   | 8763   |
| 12                                     | 150 | 0     | Self Medium   | lgK:-2.69(3SF)    | 2 | EFE   | 31910  |
| 13                                     | 160 | 0     | Self Medium   | lgK:-2.60(3SF)    | 4 | MSL   | 31908  |
| 14                                     | 190 | 0     | Self Medium   | lgK:-2.35(0.2SD)  | 4 | MSL   | 31908  |
| 15                                     | 200 | 0     | Self Medium   | lgK:-2.76(0.10SD) | 5 | MSL   | 8763   |
| 16                                     | 200 | 0     | Self Medium   | lgK:-2.43(3SF)    | 2 | EFE   | 31910  |
| 17                                     | 250 | 0     | Self Medium   | lgK:-2.16(0.10SD) | 5 | MSL   | 8763   |
| 18                                     | 250 | 0     | Self Medium   | lgK:-1.96(0.1SD)  | 4 | MSL   | 31908  |
| 19                                     | 250 | 0     | Self Medium   | lgK:-1.99(3SF)    | 2 | EFE   | 31910  |
| 20                                     | 300 | 0     | Self Medium   | lgK:-1.78(0.07SD) | 5 | MSL   | 8763   |

|    |     |   |             |                    |   |     |       |
|----|-----|---|-------------|--------------------|---|-----|-------|
| 21 | 300 | 0 | Self Medium | lgK:-1.77 (0.07SD) | 4 | MSL | 31908 |
| 22 | 300 | 0 | Self Medium | lgK:-1.69 (3SF)    | 2 | EFE | 31910 |
| 23 | 350 | 0 | Self Medium | lgK:-1.68 (0.05SD) | 5 | MSL | 8763  |

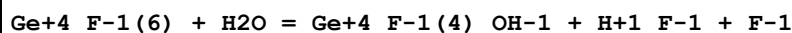
## Reaction No. 80902



Note: Ge+4 is deprecated because does not exist in aqueous solution

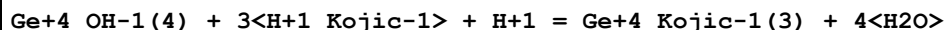
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0     | Unknown | lgK:7.24 (3SF) | 0 | MSE   | 31904  |

## Reaction No. 80903



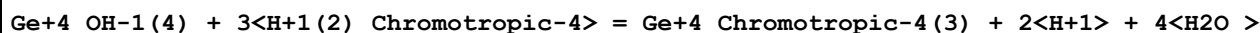
| No | t  | I-Str | Medium      | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------|-----------------|---|-------|--------|
| 1  | 0  | 0     | Self Medium | lgK:-5.92 (3SF) | 2 | MAN   | 31889  |
| 2  | 25 | 0     | Self Medium | lgK:-5.28 (3SF) | 2 | MAN   | 31889  |
| 3  | 25 | 0     | Self Medium | dH:9.5 (2SF)    | 2 | MTD   | 31889  |

## Reaction No. 80904



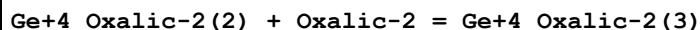
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:6.0 (2SF) | 3 | MGL   | 31905  |

## Reaction No. 80905



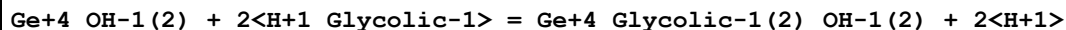
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:4.8 (2SF) | 3 | MGL   | 31905  |

## Reaction No. 80906



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0     | Unknown | lgK:3.5 (2SF) | 1 | MIX   | 32196  |

## Reaction No. 80907



Note: Ge(OH)2 deprecated, existence very much doubtful, even more in these experiments

| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:0.25 (2SF) | 0 | MGL   | 7760   |

Note (1): Complex in article:  $\text{H}_2[\text{Ge}(\text{OH})_2\text{L}_2\text{H}(-2)]$ ; simplified here:  $\text{Ge}(\text{OH})_2\text{L}_2$

## Reaction No. 80908



Note:  $\text{Ge}(\text{OH})_2$  deprecated, existence very much doubtful, even more in these experiments

| No | t  | I-Str | Medium | Value (Dev)                | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | $\lg K: 0.46 (2\text{SF})$ | 0 | MGL   | 7760   |

Note (1): Complex in article:  $\text{H}_2[\text{Ge}(\text{OH})_2\text{L}_2\text{H}(-2)]$ ; simplified here:  $\text{Ge}(\text{OH})_2\text{L}_2$

## Reaction No. 80909

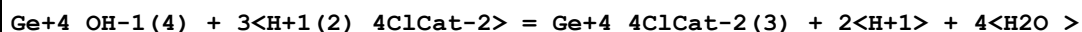


Note:  $\text{Ge}(\text{OH})_2$  is deprecated, existence very much doubtful, even more in these experiments

| No | t  | I-Str | Medium | Value (Dev)                | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaNO3  | $\lg K: 2.01 (3\text{SF})$ | 0 | MSP   | 31901  |

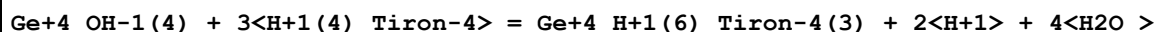
Note (1): Complex as  $\text{H}_2[\text{Ge}(\text{OH})_2(\text{CitricH}_2\text{H-1})_2]$  in publication

## Reaction No. 80910



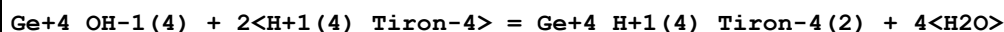
| No | t  | I-Str | Medium | Value (Dev)                | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | $\lg K: 0.65 (2\text{SF})$ | 4 | MSP   | 31872  |

## Reaction No. 80911



| No | t  | I-Str | Medium | Value (Dev)                | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------------|---|-------|--------|
| 1  | 25 | 0.11  | NaCl   | $\lg K: 2.30 (3\text{SF})$ | 0 | MSP   | 31872  |
| 2  | 25 | 0.26  | NaCl   | $\lg K: 3.10 (3\text{SF})$ | 0 | MSP   | 31872  |
| 3  | 25 | 0.5   | NaCl   | $\lg K: 3.50 (3\text{SF})$ | 3 | MSP   | 31872  |
| 4  | 25 | 1     | NaCl   | $\lg K: 3.70 (3\text{SF})$ | 3 | MSP   | 31872  |

## Reaction No. 80912



| No | t  | I-Str | Medium | Value (Dev)                | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaCl   | $\lg K: 3.88 (3\text{SF})$ | 3 | MSP   | 31872  |
| 2  | 25 | 1     | NaCl   | $\lg K: 3.90 (3\text{SF})$ | 3 | MSP   | 31872  |

## Reaction No. 80913



| Ge+4_OH-1(4) + 2<H+1_6IKoj-1> = Ge+4_6IKoj-1(2)_OH-1(2) + 2<H2O> |    |       |        |               |   |       |        |
|------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| No                                                               | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                | 25 | 0.5   | NaCl   | lgK:2.49(3SF) | 4 | MSP   | 31976  |

| Reaction No. 80914                                                   |    |       |        |               |   |       |        |
|----------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1_Comenic-2> = Ge+4_OH-1(2)_Comenic-2(2) + 2<H2O> |    |       |        |               |   |       |        |
| No                                                                   | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0.5   | NaCl   | lgK:2.25(3SF) | 4 | MSP   | 31976  |

| Reaction No. 80915                                                          |    |       |        |              |   |       |        |
|-----------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1_8OH2MeQuin-1> = Ge+4_8OH2MeQuin-1(2)_OH-1(2) + 2<H2O > |    |       |        |              |   |       |        |
| No                                                                          | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0.5   | NaCl   | lgK:3.4(2SF) | 4 | MSP   | 31977  |

| Reaction No. 80916                                                                        |    |       |        |                 |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + Glycerol = Ge+4_H+1(-4)_Glycerol_OH-1 + H+1 + 3<H2O>                       |    |       |        |                 |   |       |        |
| No                                                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0.1   | KCl    | lgK:-7.320(4SF) | 3 | MQH   | 31949  |
| Note (1): Values for react. with Ge(OH)4 and GeO(OH)3 given (K 2.69E-09); first preferred |    |       |        |                 |   |       |        |

| Reaction No. 80917                                                                       |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<Glycerol> = Ge+4_H+1(-4)_Glycerol(2)_OH-1 + H+1 + 3<H2O >               |    |       |        |                 |   |       |        |
| No                                                                                       | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                        | 25 | 0.1   | KCl    | lgK:-6.636(4SF) | 3 | MQH   | 31949  |
| Note (1): Values for react. with Ge(OH)4 and GeO(OH)3 given (K 2.63E-9); first preferred |    |       |        |                 |   |       |        |
| 2                                                                                        | 25 | 0.1   | KNO3   | lgK:-7.15(3SF)  | 3 | MSP   | 31906  |
| Note (2): NDP = 2 for these authors; reaction adapted                                    |    |       |        |                 |   |       |        |

| Reaction No. 80918                                                                       |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<EtGlycol> = Ge+4_H+1(-4)_EtGlycol(2)_OH-1 + H+1 + 3<H2O >               |    |       |        |                 |   |       |        |
| No                                                                                       | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                        | 25 | 0.1   | KCl    | lgK:-8.954(4SF) | 3 | MQH   | 31873  |
| Note (1): Values for react. with Ge(OH)4 and GeO(OH)3 given (K 2.63E-9); first preferred |    |       |        |                 |   |       |        |

|                                                                     |  |  |  |  |  |  |  |
|---------------------------------------------------------------------|--|--|--|--|--|--|--|
| Reaction No. 80919                                                  |  |  |  |  |  |  |  |
| Ge+4_OH-1(4) + EtGlycol = Ge+4_H+1(-4)_EtGlycol_OH-1 + H+1 + 3<H2O> |  |  |  |  |  |  |  |

| No                                                                                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 1                                                                                                                | 25 | 0.1   | KCl    | lgK:-8.140(4SF) | 3 | MQH   | 31873  |
| Note (1): Values for react. with Ge(OH) <sub>4</sub> and GeO(OH) <sub>3</sub> given (K 2.63E-9); first preferred |    |       |        |                 |   |       |        |

| Reaction No. 80920                                                                                               |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 12PrDiol = Ge+4_H+1(-4)_12PrDiol_OH-1 + H+1 + 3<H2O>                                              |    |       |        |                 |   |       |        |
| No                                                                                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                | 25 | 0.1   | KCl    | lgK:-8.298(4SF) | 3 | MQH   | 31873  |
| Note (1): Values for react. with Ge(OH) <sub>4</sub> and GeO(OH) <sub>3</sub> given (K 2.63E-9); first preferred |    |       |        |                 |   |       |        |

| Reaction No. 80921                                                                                                 |    |       |        |                 |   |       |        |
|--------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<12PrDiol> = Ge+4_H+1(-4)_12PrDiol(2)_OH-1 + H+1 + 3<H2O >                                         |    |       |        |                 |   |       |        |
| No                                                                                                                 | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                  | 25 | 0.1   | KCl    | lgK:-8.521(4SF) | 3 | MQH   | 31873  |
| Note (1): Values for react. with Ge(OH) <sub>4</sub> and GeO(OH) <sub>3</sub> given (Ksp 2.63E-9); first preferred |    |       |        |                 |   |       |        |

| Reaction No. 80922                                                         |    |       |               |               |   |       |        |
|----------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1_Lactic-1> = GeO+2_H+1(-1)_Lactic-1(2) + H+1 + 3<H2O > |    |       |               |               |   |       |        |
| No                                                                         | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                          | 18 | 0     | Inf. Dilution | lgK:2.0(2SF)  | 2 | MGL   | 31980  |
| Note (1): Complex in publication GeOL2HH-2, simplified here                |    |       |               |               |   |       |        |
| 2                                                                          | 18 | 1     | NaCl          | lgK:0.37(2SF) | 2 | MCN   | 31980  |
| Note (2): Complex in publication: GeOL2HH-2, simplified here               |    |       |               |               |   |       |        |

| Reaction No. 80923                                                                                       |    |       |               |               |   |       |        |
|----------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<H+1(2)_Malic-2> = GeO+2_H+1_Malic-2(2) + H+1 + 3<H2O>                                   |    |       |               |               |   |       |        |
| No                                                                                                       | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                        | 18 | 0     | Inf. Dilution | lgK:2.9(2SF)  | 2 | MGL   | 31980  |
| Note (1): Complex in publication: GeOL2HH-2,with NDP of ligand 1; with NDP 2 GeOL2H3H-2, simplified here |    |       |               |               |   |       |        |
| 2                                                                                                        | 18 | 1     | NaCl          | lgK:0.68(2SF) | 2 | MCN   | 31980  |
| Note (2): Complex in publication: GeOL2HH-2,with NDP of ligand 1; with NDP 2 GeOL2H3H-2, simplified here |    |       |               |               |   |       |        |

|                                                                   |  |  |  |  |  |  |  |
|-------------------------------------------------------------------|--|--|--|--|--|--|--|
| Reaction No. 80924                                                |  |  |  |  |  |  |  |
| Ge+4_OH-1(4) + H+1_Tartaric-2 = GeO+2_H+1(-1)_Tartaric-2 + 3<H2O> |  |  |  |  |  |  |  |

| No                                           | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
|----------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| 1                                            | 18 | 1     | NaCl   | lgK:5.1(2SF) | 2 | MCN   | 31980  |
| Note (1): Complex as GeOLHH-2 in publication |    |       |        |              |   |       |        |

| Reaction No. 80925                                                           |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<DMannitol> = Ge+4_H+1(-4)_DMannitol(2)_OH-1 + H+1 + 3<H2O > |    |       |        |                |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                            | 20 | 0.1   | KNO3   | lgK:-4.0(2SF)  | 4 | MGL   | 31905  |
| 2                                                                            | 25 | 0.1   | KCl    | lgK:-4.05(3SF) | 4 | MGL   | 31949  |

| Reaction No. 80926                                                           |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<DSorbitol> = Ge+4_H+1(-4)_DSorbitol(2)_OH-1 + H+1 + 3<H2O > |    |       |        |                |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                            | 20 | 0.1   | KNO3   | lgK:-3.7(2SF)  | 4 | MGL   | 31905  |
| 2                                                                            | 25 | 0.1   | KCl    | lgK:-3.49(3SF) | 4 | MGL   | 31876  |

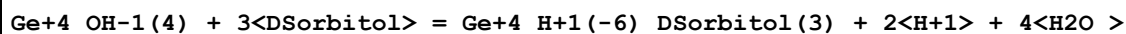
| Reaction No. 80927                                                         |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<DGlucose> = Ge+4_H+1(-4)_DGlucose(2)_OH-1 + H+1 + 3<H2O > |    |       |        |                 |   |       |        |
| No                                                                         | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                          | 20 | 0.1   | KNO3   | lgK:-6.33(3SF)  | 0 | MGL   | 31906  |
| 2                                                                          | 25 | 0.1   | KCl    | lgK:-5.120(4SF) | 2 | MGL   | 31873  |

| Reaction No. 80928                                                    |    |       |               |                 |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 3<H+1(2)_Tiron-4> = Ge+4_Tiron-4(3) + 2<H+1> + 4<H2O > |    |       |               |                 |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0     | Inf. Dilution | lgK:-2.307(4SF) | 0 | MGL   | 31891  |
| 2                                                                     | 25 | 0.294 | KCl           | lgK:1.829(4SF)  | 0 | MGL   | 31891  |
| 3                                                                     | 25 | 0.368 | KCl           | lgK:2.521(2SF)  | 3 | MGL   | 31891  |
| 4                                                                     | 25 | 0.483 | KCl           | lgK:3.244(2SF)  | 3 | MGL   | 31891  |
| 5                                                                     | 25 | 0.617 | KCl           | lgK:3.753(2SF)  | 3 | MGL   | 31891  |
| 6                                                                     | 25 | 0.726 | KCl           | lgK:4.037(2SF)  | 3 | MGL   | 31891  |
| 7                                                                     | 25 | 0.791 | KCl           | lgK:4.192(4SF)  | 3 | MGL   | 31891  |
| 8                                                                     | 25 | 1.009 | KCl           | lgK:4.504(2SF)  | 3 | MGL   | 31891  |
| 9                                                                     | 25 | 1.572 | KCl           | lgK:4.711(2SF)  | 3 | MGL   | 31891  |

| Reaction No. 80929                                                           |  |  |  |  |  |  |  |
|------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| Ge+4_OH-1(4) + 2<LRhamnose> = Ge+4_H+1(-4)_LRhamnose(2)_OH-1 + H+1 + 3<H2O > |  |  |  |  |  |  |  |

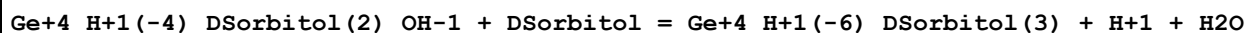
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:-5.340 (2SF) | 4 | MGL   | 31981  |

## Reaction No. 80930



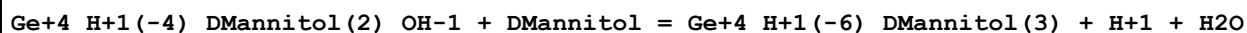
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:-12.3 (3SF) | 4 | MGL   | 31905  |

## Reaction No. 80931



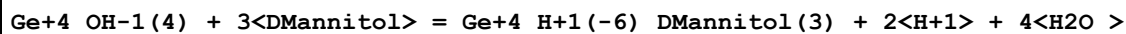
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:-8.6 (2SF) | 4 | MGL   | 31905  |

## Reaction No. 80932



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:-9.7 (2SF) | 4 | MGL   | 31905  |

## Reaction No. 80933



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:-13.7 (3SF) | 4 | MGL   | 31905  |

## Reaction No. 80934



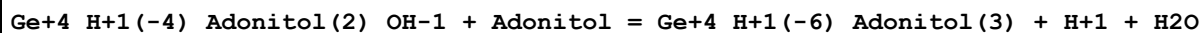
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:-10.6 (3SF) | 4 | MGL   | 31906  |

## Reaction No. 80935



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:-10.0 (3SF) | 4 | MGL   | 31906  |

## Reaction No. 80936



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:-10.6 (3SF) | 4 | MGL   | 31906  |

| Reaction No. 80937                                                         |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ge+4_OH-1(4) + 2<Adonitol> = Ge+4_H+1(-4)_Adonitol(2)_OH-1 + H+1 + 3<H2O > |    |       |        |                 |   |       |        |
| No                                                                         | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                          | 20 | 0.1   | KNO3   | lgK:-5.41 (3SF) | 4 | MGL   | 31906  |

### Data Assessment

| Weight | Assessment                                    |
|--------|-----------------------------------------------|
| 9      | highest accuracy                              |
| 8      | multi-determined; excellent dependability     |
| 7      | trustworthy; outstanding single determination |
| 6      | better than average reliability               |
| 5      | average single determination                  |
| 4      | some doubt                                    |
| 3      | poorly known                                  |
| 2      | guess to achieve consistency (or just as bad) |
| 1      | no information but consistent                 |
| 0      | problematic/inconsistent                      |

### Techniques

| Abbrev | Method                          |
|--------|---------------------------------|
| CRV    | Critical review                 |
| EAE    | Automated extrapolation         |
| EDH    | Debye-Hueckel corrections       |
| EES    | Personal estimate               |
| EFE    | Free Energy relations           |
| ELC    | Linear Combination of Reactions |
| ENS    | Not specified                   |
| EOD    | Calculated from data of others  |
| ESC    | Standard conditions anchor      |
| MAN    | Chemical analysis               |
| MAX    | Anion exchange                  |

|     |                                      |
|-----|--------------------------------------|
| MCN | Conductivity                         |
| MCO | Colorimetry (often for measuring pH) |
| MDG | Combination of Thermodynamic Data    |
| MEF | Electromotoric force, not specified  |
| MFP | Freezing point                       |
| MGL | Glass electrode                      |
| MHY | Emf with hydrogen electrode          |
| MIX | Ion exchange                         |
| MPL | Polarography                         |
| MPT | Potentiometry                        |
| MQH | Quinhydrone electrode                |
| MSC | Miscellaneous                        |
| MSE | Solvent Extraction                   |
| MSL | Solubility                           |
| MSP | Spectroscopy                         |
| MTD | Temperature difference               |

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